



The post-2030 baseline decline rate: starting point, adjustments and verification

A mechanical calibration using the published facility register gives a starting point of 3.53% for the 62% target and 5.99% for the 70% target under current rules

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What the review has to decide. The consultation paper asks that the post-2030 baseline decline rate be calibrated to the adopted 2035 target^[1]. The decision involves two linked parameters: the default rate that applies to most facilities, and the decline path that applies to the 21 trade-exposed baseline-adjusted (TEBA) facilities. Published estimates provide a useful range for the final rate. This submission proposes a calibration process that can be independently traced and reproduced. It first establishes a mechanical starting point using public register data, published commodity projections and a limited set of explicit scenario assumptions. It then introduces the additional policy and behavioural assumptions separately and uses subsequent annual data to test whether those assumptions are borne out.

Core finding. The default decline rate and the post-2030 TEBA pathway must be calibrated together. For a given 2035 target, reducing the TEBA decline rate by one percentage point per year increases the default rate required for the other 187 facilities by approximately 0.2 percentage points per year. Every combination that meets the target lies on one iso-target curve. The policy decision therefore amounts to selecting a point on that curve.

The mechanical starting point. Under current rules the two rates are equal, and the starting point is in the table below. It is a mechanical starting point under stated assumptions. The final policy rate must also reflect forward-looking adjustments, including the reserve and expected on-site abatement.

2035 target	Default rate, non-TEBA	TEBA post-2030 rate
62% (point target 68.8 Mt)	3.53%	3.53%
70% (point target 54.3 Mt)	5.99%	5.99%

Recommendation

1 Adopt and publish a transparent, reproducible framework for post-2030 calibration, comprising three sequential steps and subject to periodic review

Step one — Establish the starting point. Fix the starting point for the default rate and the TEBA rate jointly, either recomputed by the Department on the same assumptions or taken from the mechanical starting point in this submission. The 3.53% and 5.99% in this submission correspond to this step.

Step two — Quantify the adjustments. Quantify and publish each ex-ante adjustment separately, for production, entry and exit, TEBA treatment, on-site abatement, below-threshold credits and the reserve, stating how much each moves the rate. Large publicly identified projects should be listed individually rather than incorporated only through an aggregate growth allowance.

Step three — Monitor outcomes and recalibrate where necessary. Test the material assumptions against subsequent annual data, by publishing the relative-intensity effect of covered facilities with each annual register, comparing the results with the abatement pathway used in the calibration, and setting out the recalibration mechanism in advance.

Declaration of interest. Helio vulcan provides paid research to parties with interests in Australian industrial energy assets. This submission was prepared independently and was not commissioned or funded by any third party. The full declaration is in section 1.4.

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How to read this submission. Page 1 gives the finding, the starting point and the recommendation. Sections 1 to 7 set out the calibration framework, the starting point, the adjustments, the comparison with DCCEEW, the Project Ares example and the monitoring method. Annexes A to H hold the equations, sources, facility lists and limitations. The accompanying workbook reproduces every table and figure with its formulas.

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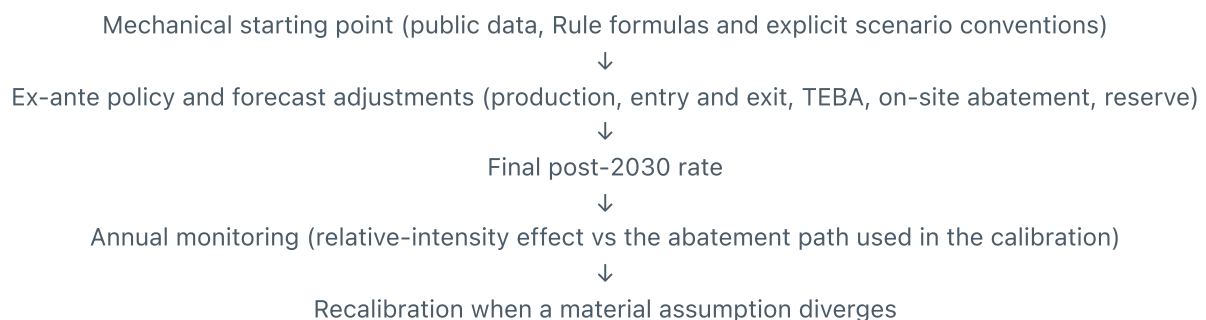
1. Purpose, scope and recommendation

1.1 The decision before the review

The consultation paper states that the decline rate for 2030-31 to 2034-35 "must be calibrated to meet the agreed target" (p 43) ^[1]. It lists six calibration factors: the target itself, production growth at existing facilities, facility entry and exit, uptake of concessional baselines, policy changes arising from this review, the reserve, and credits issued to below-threshold facilities. Section 3.3.3 of the same paper discusses a catch-up mechanism for TEBA facilities. The final rate therefore depends on a series of policy choices and forecasting inputs, each of which should be made explicit.

The published estimates, from RepuTex, EY and Australian Energy Producers, provide different ranges for the final rate. The published information on their methods generally does not allow readers to distinguish the mechanical effect of the Rule from assumptions about production, abatement, entry and exit. Public information does not currently show how each assumption changes the final rate. This submission provides that missing calibration bridge: a mechanical starting point based on public data, published commodity projections and explicit scenario conventions; the adjustments that connect it to the final rate; and a method for testing those adjustments after implementation.

1.2 The calibration framework proposed



The framework distinguishes three types of result that are often conflated. The **mechanical starting point** is calculated from public data using explicit scenario conventions. It calibrates the default and TEBA rates jointly and can be independently reproduced. A **behavioural projection** adds expectations about production, entry, exit and on-site abatement. The **final policy rate** then incorporates the reserve and other policy judgements. If the final rate is below the starting point because of expected on-site abatement, that expectation is a substantive component of the calibration. Actual outcomes should be measured annually and should trigger reassessment if they diverge materially from the assumption used in the calibration. This submission estimates the mechanical starting point, providing a reference against which the effects of behavioural forecasts and policy choices can be measured.

1.3 Scope

This submission addresses the method used to calibrate the post-2030 decline rate and the transparency of that calibration. It reports results for both the 62% and 70% endpoints of the adopted target range, and treats the TEBA pathway, the facility population and the production outlook as separately identifiable parameters.

It does not recommend a preferred point within the target range. Nor does it recommend changes to the minimum baseline rule (Safeguard Rule s 10(1): any baseline that computes below 100,000 t is set to 100,000 t, compilation p 10) ^[3], the coverage threshold (Rule s 8: 100,000 t of scope 1 emissions a year, p 9), the ACCU (Australian carbon credit unit) rules or the intensity determinations. It does not forecast the operating decisions of individual existing Safeguard facilities; section 5 presents illustrative scenarios for the proposed Project Ares based on publicly stated project specifications. It does not recommend a preferred TEBA treatment; section 2 and Annex A.9 quantify the effect of each treatment on the default rate. The main body responds to section 4.2; section 5 responds to the questions in section 4.1 on the reserve and new entrants; section 6 responds to the watch point in section 2.

The analysis has five material limitations. Production follows national commodity projections rather than facility-level plans. The FY2024-25 intensity blend is held constant, so the annual transition under Rule s 13 (the blend of a facility's own intensity and the industry best-practice intensity, moving each year toward best practice, p 12) is not modelled. Only owner-announced or completed closures are included. The ± 4 Mt figure in section 6 is an indicative sensitivity derived from a sample of coal mines, not a confidence interval. Finally, the starting point is calibrated to the FY2034-35 point target as a conservative calibration convention; the budget-based result is reported separately.

1.4 Evidence and interests

The analysis uses the Regulator's published FY2024-25 register^[4] and TEBA register^[16], the Safeguard Rule^[3], the consultation paper^[1], the Department's published projections^[5] and their methodology^[17], OCE commodity volumes^[18] and public project lists^{[10] [14] [15]}. This submission is policy analysis and does not constitute legal or financial product advice, or a recommendation concerning any company or financial product discussed. Questions of statutory interpretation are identified for DCCEEW's consideration. The algorithm in Annex A and the accompanying workbook allow third parties to trace the calculation facility by facility and confirm the reported rates by entering them into the accompanying model; the rates themselves come from the bisection procedure described in Annex A.7. Some facility-level working papers contain company-specific matching and classification decisions and are available to DCCEEW on request.

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This submission was prepared independently. It was not commissioned, funded or reviewed by any client. No commercial entity discussed as a case study in this submission is, or has been, a client of Heliovolcan.

2. A reproducible calibration starting point

2.1 Why the review is choosing a pair of rates

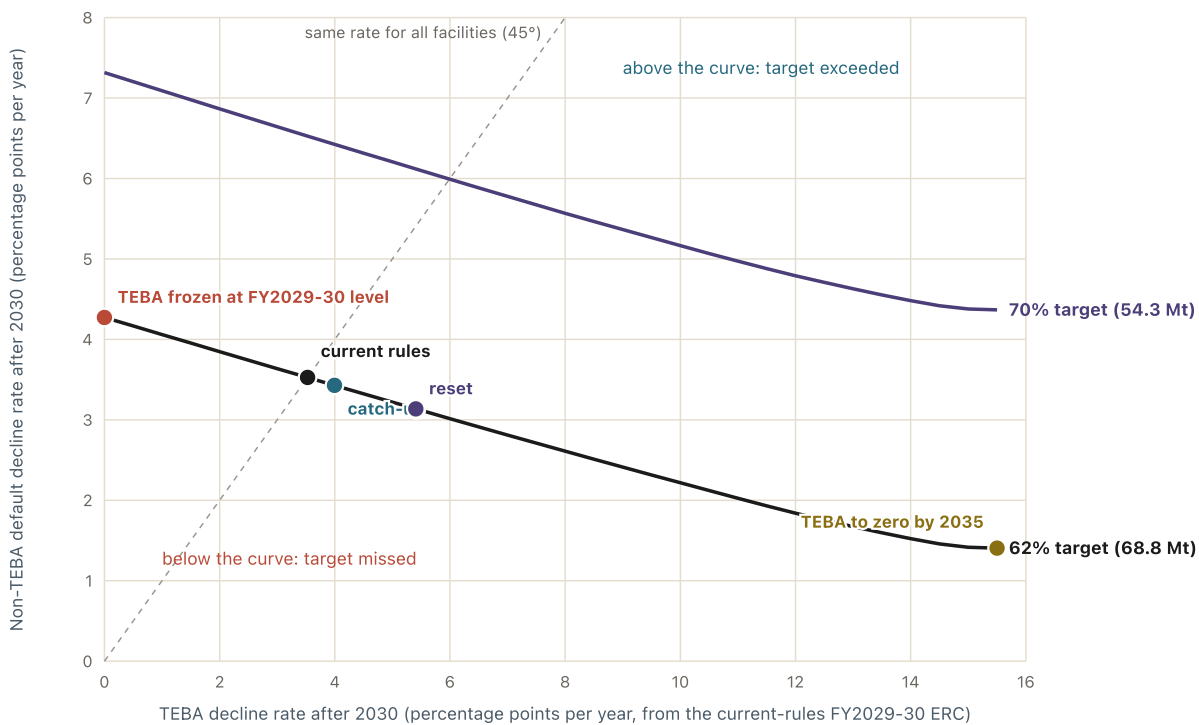
A Safeguard baseline declines each year by an "emissions reduction contribution" (ERC). Under Rule s 11, an existing facility's baseline is calculated as production multiplied by the applicable emissions-intensity value and the ERC (compilation p 10), and s 31 gives the table of default ERC values by year (p 28)^[3]. Most facilities follow the same default ERC path: from 0.951 in FY2023-24, down 4.9 points a year to 0.657 in FY2029-30, then down d points a year after 2030. That d is the number to be set.

The 21 facilities with a TEBA determination follow a different path. A TEBA facility has been found by the Regulator under Rule s 42 to be trade-exposed and is permitted a slower baseline decline (p 36; the facilities are listed in the register^[16]). While the concessional baseline applies, the facility's ERC declines more slowly. After the determination expires, it follows the default slope from a higher starting ERC. By FY2034-35 the TEBA facilities therefore retain about 2.41 Mt more baseline than the default path would give. For a fixed 2035 target, the additional TEBA baseline must be offset by a lower aggregate baseline for the remaining facilities. Moving from the current-rules treatment to reset removes approximately 2.41 Mt of TEBA baseline in FY2034-35 and reduces the required default rate by 0.39 percentage points; across that policy change, the average effect is therefore approximately 0.16 percentage points of default decline per 1 Mt of TEBA baseline removed. The post-2030 TEBA pathway and the default rate should therefore be calibrated together.

2.2 The iso-target curve

Figure 2.1 shows the feasible combinations of the two decline rates. The horizontal axis is the post-2030 decline rate for TEBA facilities, and the vertical axis is the default rate for the other 187 facilities. Every point on a curve meets the same FY2034-35 target. Combinations below the curve miss the target; combinations above it exceed the target. Mathematically, each curve is a level set of FY2034-35 compliant net emissions as a function of the two rates. Annex A.9 provides the formal definition and derives the slope.

Figure 2.1 — Combinations of TEBA and non-TEBA decline rates that meet the FY2034-35 target (starting-point population)



Below the curve the target is missed; above it the target is exceeded. Current rules sit where the curve meets the 45° line; catch-up and reset sit to its right. The slope is approximately -0.2 : each additional percentage point of TEBA decline reduces the required default rate by approximately 0.2 percentage points. Coordinates of the marked points (TEBA rate, default rate): current rules (3.53, 3.53); catch-up $\approx$ (4.00, 3.43); reset $\approx$ (5.4, 3.14); TEBA frozen at the FY2029-30 level (0, 4.27); TEBA to zero by 2035 (15.5, 1.41).

Three policy-relevant points matter most. Under current rules TEBA facilities decline at the same rate as the others from a higher starting level after expiry, so they sit where the 45° line meets the curve, at (3.53, 3.53). The catch-up mechanism in section 3.3.3 of the consultation paper brings TEBA facilities back onto the default path over 20 years from 2031 and corresponds to about (4.00, 3.43). The reset option in section 3.3.4 moves them straight onto the default ERC from 2031 and corresponds to about (5.4, 3.14). The slope of approximately -0.2 reflects the relative size of the two constrained facility groups at the current-rules solution. Benchmark-weighted output is about 21 Mt for TEBA facilities and about 97 Mt for the other constrained facilities. Their ratio is approximately 0.21, so a one-percentage-point increase in the post-2030 TEBA decline rate reduces the default rate required for the other facilities by approximately 0.21 percentage points, holding the target constant. Annex A.9 reports two additional cases: freezing TEBA ERCs at their FY2029-30 levels and reducing TEBA ERCs to zero by FY2034-35.

2.3 Assumptions used for the starting point

The starting-point calculation assigns an explicit value or treatment to each of the six calibration inputs. Most retain the current settings or make no additional adjustment. Table 2.1 lists how each is treated in the starting point and whether it has to be adjusted again for the final rate. The basis for each, and its effect on the required rate under alternative assumptions, are in Annex E, Table E.2.

Table 2.1 — Treatment of the six calibration inputs in the starting point

Calibration input	Treatment in the starting point	For the final rate
2035 target	Both the 62% and 70% targets; share 29.8%; the compliance condition is the FY2034-35 point target (68.8 / 54.3 Mt, consultation paper p 41) ^[1] as a conservative calibration convention	Replaced once the target is set; the budget-basis solution is in Annex A, Table A.2

Calibration input	Treatment in the starting point	For the final rate
Production at existing facilities	Each facility's output index follows the OCE national volume for its commodity ^[18] ; growth above the base year is assigned to new entrants and existing facilities are not scaled up; after 2031 thermal coal follows the IEA ^[19] and the rest is flat	Replaced by the Department's own production projections
Entry and exit	All 62 publicly visible pipeline projects ^{[10] [14] [15]} counted in full, without probability weighting, phased in using the commissioning-year assumptions described in Annex A.5; exit counts only the seven owner-announced or completed closures	Adjusted for probability and timing; pending closures and curtailments added separately
TEBA	The 21 current determinations follow the register ^[16] to expiry, then decline at the default slope from the higher level, at the same rate as non-TEBA facilities after 2030	Adjusted to the final TEBA policy (catch-up, reset, renewal)
On-site abatement	None; every facility complies in full every year, net = min(covered, baseline)	A separate adjustment, monitored annually (section 6)
Reserve, below-threshold credits, other policy changes	None	Separate adjustments

The facility count follows the Regulator, 208 facilities^[4]. The published register has 228 rows. Removing the 19 facilities that are eligible facilities under Rule s 58B (once above the threshold, now below it but still able to earn credits, compilation p 48)^[3] leaves 209 rows. Telfer Gold Mine occupies two of them because of a mid-year change of operator. After the two Telfer rows are consolidated, the dataset contains 208 facilities with total covered emissions of 132.8 Mt. The calculation proceeds in four steps. First, baseline divided by ERC gives the benchmark-weighted output indicator *J*. Second, *J* multiplied by the applicable ERC gives the annual baseline. Third, modelled compliant net emissions are calculated as the smaller of covered emissions and the applicable baseline. Finally, bisection identifies the decline rate at which FY2034-35 net emissions equal the target. The full statement and equations are in Annex A.

2.4 Results

Table 2.2 — Default rate under three TEBA treatments, starting-point population, percentage points a year

TEBA treatment	Default rate, 62% target	Default rate, 70% target
Current rules: decline from the higher level after expiry at the same rate (starting point)	3.53	5.99
Catch-up: 20-year transition proportion from FY2030-31 (consultation paper option)	3.43	5.89
Reset: default ERC from FY2030-31	3.14	5.60

Rates are expressed as linear percentage-point reductions in the original baseline. The legislated indicative rate of 3.285% is below every result in the table. The renewal-to-FY2029-30 sensitivity is 4.03 / 6.50. The full table of three populations by five TEBA treatments is Annex A, Table A.1.

Table 2.3 — Sensitivity of the starting point to the facility population, current rules, percentage points a year

Scenario	62% target	70% target
Existing 208 facilities only, production calibrated to OCE	3.46	5.96
Plus the publicly visible pipeline, counted in full without probability weighting	3.67	6.13
Less seven owner-announced or completed closures (starting point)	3.53	5.99

All three rows calibrate production to OCE commodity volumes and solve to the FY2034-35 point target. Three comparison cases are in Annex A, Table A.2: the reference case with production flat at FY2024-25, tracker expiry years all treated as closures, and solving to the five-year budget instead of the point target. The point target is the more stringent of the two constraints and is therefore used for the starting-point calibration. This is a conservative modelling convention, not a conclusion about the legally required interpretation.

3.53% and 5.99% correspond to the current TEBA treatment; a different TEBA treatment changes the matching default rate. These results are a mechanical starting point under stated assumptions, reproducible by any third party. The final policy rate must still be determined after incorporating the adjustments discussed in section 3.

3. Adjustments required to derive the final rate

Section 2 established a mechanical basis that does not incorporate facility-level behavioural forecasts or additional on-site abatement. Four classes of adjustment separate it from the final rate: the facility population and production, TEBA policy, on-site abatement, and the reserve and other policy settings. Where public data supports an adjustment, its magnitude is given below. Where an adjustment depends on non-public facility information, the submission identifies the agency or other party that holds the relevant data.

3.1 Production and the facility population

Three factors materially affect the production and facility-population assumptions.

First, the visible pipeline. The 62 projects with a capacity figure in public trackers (41 coal mines, 5 LNG trains, 16 iron ore mines)^{[10] [14] [15]} would add 17.0 Mt a year of covered emissions once all operate, with baselines of 2.59 Mt at the Rule's own best-practice intensities. The starting-point model phases them in using the commissioning profile described in Annex A.5, which raises the rate needed for the 62% target by about 0.21 of a point. This is an upper-bound convention in which all identified capacity enters. DCCEEW should replace it with a project-level forecast that reflects both commissioning timing and the probability of entry.

Second, confirmed closures. The seven owner-announced or completed closures (three coal mines and four other facilities, 2.03 Mt covered) reduce the required rate by approximately 0.14 percentage points. A further seven coal mines have closure years recorded in the tracker, but those dates generally refer to approval or lease expiry rather than confirmed closure, and most have extensions under application. Treating all of them as closures would lower the rate by a further 0.19 of a point. The starting point therefore includes only owner-announced or completed closures and reports the expiry-year convention as a sensitivity.

Third, production. The starting point calibrates to OCE national volumes by commodity^[18]. Holding production flat at FY2024-25 instead (the reference case in Annex A, Table A.2) raises the rate by 0.37 of a point, mostly from thermal coal, LNG, alumina and east-coast gas. Two items are known in direction but not included in the starting point. Rio Tinto has announced a 40% reduction in production at Yarwun from October 2026; incorporating that reduction would lower the required rate by approximately 0.08 percentage points. Whyalla, Nyrstar, the Bell Bay smelter, Mount Isa smelting, two refineries and Longford total about 5.7 Mt of pending decisions; several of these facilities are individually comparable in scale to, or larger than, the confirmed coal closures combined. The 62 projects, the closure list and the production factors are in Annex D.

3.2 The TEBA policy adjustment

The curve in section 2 already quantifies this item. Moving from current rules to catch-up lowers the default rate by 0.10 of a point, moving to reset lowers it by 0.39, and renewing every determination to FY2029-30 raises it by 0.50. The TEBA treatment is therefore a separate, quantifiable adjustment between the mechanical starting point and the final rate, and the decision document should publish the chosen TEBA path together with its effect on the default rate. Annex A.9 sets out the ERC path under each treatment; under current rules, TEBA facilities retain about 2.41 Mt more baseline in FY2034-35 than the default path would give.

3.3 The on-site abatement adjustment

The starting point counts no future on-site abatement. The Department's projection does, rising each year from FY2025-26 to a cumulative 18.55 Mt by FY2029-30 and 35.23 Mt by FY2034-35^[6]. This item is the main source of the post-2030 difference between the starting point and the Department's projection (section 4). If the final rate is below the mechanical starting point, the share attributable to this assumption should be disclosed explicitly. Section 6 proposes an annual test of this assumption using the published facility register.

3.4 The reserve and other adjustments

The reserve. The consultation paper describes the reserve as a buffer for uncertainty in production growth and the uptake of concessional baselines (p 7). In the settings adopted for 2030, it allowed for emissions associated with production growth of up to 0.9% a year (p 9)^[1]. The final calibration should state the size of the reserve, the assumptions it covers and the effect it has on the decline rate.

Inputs not available in the public record. Several inputs needed for the final calibration can be quantified only by DCCEEW or the Regulator. These include future TEBA applications and renewals after 2030; SMCs issued to facilities that fall below the coverage threshold (Safeguard Mechanism credit units, issued to a facility whose emissions are below its baseline and transferable to other facilities for compliance); the annual transition of the Rule s 13 intensity blend (standard 20:80 to 100:0, coal 10:90 to 50:50), whose effects may differ materially across facilities even if the aggregate effect is small; and other policy changes arising from this review. These inputs should be reported separately so that their effects are not absorbed into an unexplained aggregate reserve.

Matters requiring clarification. First, Rule s 57 (compilation pp 45–46)^[3] defines BEN (baseline emissions number) as the baseline that would apply "if subsection 10(1) had not been made". DCCEEW should clarify whether the minimum baseline reduces a facility's compliance obligation without increasing the SMCs it can generate. Second, the consultation paper reports that concessional baselines had an effect approximately 0.4 Mt greater than expected (p 10). The final calibration should disclose the assumed future uptake of concessional baselines and explain how that variance is reflected in the reserve or in the final decline rate. Large publicly identifiable projects should likewise be listed separately, with the project-level amounts reconciled to the new-entrant component of the reserve; section 5 illustrates this approach. Table 3.1 brings the resulting adjustments together, and completing and publishing this bridge would make the final calibration independently traceable:

Table 3.1 — Calibration bridge from the mechanical starting point to the final rate, 62% target, percentage points a year

Stage	Quantified from public data	Final value
Mechanical starting point (current rules)	3.53	3.53
Final TEBA treatment	Catch-up –0.10; reset –0.39; renewal +0.50	Departmental input required
Department's production projection replacing the OCE calibration	Reference case (flat production) +0.37; the Department's basis is not public	Departmental input required
Probability- and timing-adjusted entry and exit	All entry +0.21; confirmed closures –0.14; about 5.7 Mt pending not counted	Departmental input required
On-site abatement assumption	None; the Department's path reaches 18.55 Mt cumulative by FY2029-30	Departmental input required
Reserve, below-threshold credits, TEBA renewals and new applications	None; renewal sensitivity +0.50	Departmental input required
Final post-2030 rate	—	Departmental input required

The same table for the 70% target starts from 5.99. With each row stating how much it moves the rate, a third party can rebuild the final rate from the starting point.

4. Comparison with DCCEEW's published projections

DCCEEW's published projections provide an external check on both the scale of the starting point and its treatment of the Rule. The comparison applies the same mechanical relationships over the years covered by DCCEEW's projections and measures the degree of agreement between the two series.

4.1 Broad agreement before 2030

For 2030 the Department reports gross Safeguard emissions of 122 Mt CO₂-e, net unit demand of –26 Mt (the net surrender of ACCUs and SMCs by covered facilities) and net Safeguard emissions of 90 Mt^[5]. Subtracting net unit demand from gross emissions gives 96 Mt, 6 Mt above the reported net figure of 90 Mt; the difference is SMC generation by facilities below their baselines, which the report presents separately. Adding SMC generation back to the reported net figure reconciles the Department's published years with the modelled compliant-net basis used here (Annex B.1). Three comparisons are particularly informative. The starting-point decade total is 1,171.8 Mt, compared with DCCEEW's 1,181 Mt^[5], a difference of 0.8%. The corresponding FY2029–30 aggregate baselines are 93.04 Mt and 95.94 Mt^[6]. FY2034–35 reference emissions before abatement are 142.5 Mt and 141.8 Mt respectively^[6], with a ten-year mean difference of 1.1%.

Table 4.1 — Comparison of the starting-point results with the Department's published series, Mt CO₂-e (TEBA under current rules)

Quantity	DCCEEW ^[5] ^[6]	This analysis, starting point after confirmed closures	This analysis, existing facilities	This analysis, plus new entrants
Decade budget NGER Act s 3(2)(b): total net FY2020–21 to FY2029–30 (printed p 2) ^[2]	1,181	1,171.8	1,169.2	1,174.7
Aggregate baselines, FY2029–30	95.94	93.04	92.87	94.15
Net emissions, FY2029–30 against the legislated 100 Mt point target	90	91.21	91.04	92.32
Net emissions, FY2034–35 Department: behavioural projection; this analysis: legislated 3.285%	66	70.25	69.83	71.11
Reference emissions, FY2034–35 before any on-site abatement; the Department's includes production change	141.81	142.5	127.3	144.3

†. TEBA under current rules (assumption six). The Department's reference emissions are taken from the chart data for Figure 14^[6].

4.2 The post-2030 difference comes from explicit assumptions

At the legislated 3.285% (the value of the s 31 default ERC table for each year from 2030–31, compilation p 28)^[3], the starting point reaches FY2034–35 net emissions of 70.25 Mt, 1.4 Mt above the 62% target of 68.8 Mt^[1]. The Department's projection of 66 Mt is below the target. Because the two calculations apply broadly comparable treatments of the Rule, the remaining difference appears to arise mainly from forward-looking assumptions. The Department assumes about 3% a year of on-site abatement and production change^[17]; the starting-point model assumes no additional on-site abatement. Put differently, for 3.285% to reach the 62% target in 2035, abatement, production decline, uncounted closures or TEBA claw-back would collectively need to reduce the required decline rate by the remaining 0.24 percentage points, and no quantified public estimate is currently available for any of those items.

4.3 What the comparison means, and its limits

The comparison supports one limited but important conclusion: the mechanical basis of this model is broadly compatible with the one underlying DCCEEW's projections and can serve as a transparent public calibration starting point. The post-2030 differences mainly reflect different assumptions about on-site abatement, production, entry and exit and TEBA treatment. Table 2.3 shows that alternative facility-population assumptions change the required rate by approximately 0.2 percentage points, broadly comparable with the effect of the principal TEBA alternatives. The comparison is not used to assess the accuracy of the Department's behavioural projection.

4.4 Comparison with published estimates

Four published estimates are currently available, alongside forthcoming modelling commissioned by the Climate Change Authority: RepuTex at 3.5%–6.8% (existing facilities only)^[7], EY at 5%–7%^[8], Australian Energy Producers' 8%–14% submitted to the CCA^[9] and restated here to 6.7%–9.5% on the adopted target and a linear basis, and the starting point in this submission. The existing-facilities result is within 0.03 percentage points of RepuTex's estimate and 1.47 points below EY's estimate. The differences are likely to reflect assumptions about behavioural responses, the reserve and TEBA treatment, as well as whether decline rates are reported on a linear or compound basis, rather than the public data itself. On our reading of the published material, no other estimate reports the TEBA rate together with its link to the default rate. Each estimate's basis and the figure from this submission are in Annex C.

5. An example of an ex-ante adjustment: a large identifiable new entrant

Section 3 argued that the final calibration should adjust the starting point for new entrants separately. This section examines how a large off-grid, self-supplied project would affect the Safeguard calibration if it became a covered facility, and why such projects should be listed individually rather than incorporated only through an undisclosed aggregate growth allowance. In its August submission to the Senate inquiry into artificial intelligence and data centres, Heliovolcan recommended publishing a regulatory coverage map by grid-connected, non-grid and off-grid self-supplied projects^[20].

Why Ares matters. Heliovolcan checked the power source of every site on its national data-centre register as at 12 September 2026 (162 sites at that date; the register is being revised as site-level profiles are completed). The multi-gigawatt campuses in the NEM are grid-supplied with diesel backup. The only identified projects with on-site gas generation are off-grid projects in the Northern Territory. Project Ares at Murrniji is the only publicly identified project for which the available project documents expressly anticipate coverage under the Safeguard Mechanism. The project is proposed to comprise 1 GW of IT load, 3,000 MWp of solar PV, 16 GWh of battery storage and 1,038 MW of gas generation, and is designed to operate off-grid. Its EPBC referral (the environmental referral under the Environment Protection and Biodiversity Conservation Act 1999) identifies both the NGER Act and the Safeguard Mechanism as applicable Commonwealth legislation^[21]. The proponent's referral report to the Northern Territory Environment Protection Authority goes further: it states that the project "is expected to be a registered facility under the Commonwealth Safeguard Mechanism" and that applicable thresholds will be assessed and registration made before phase one commercial operation^[40]. The referral was declared a controlled action under the EPBC Act on 4 September 2026^[21], and the proponent's schedule gives a final investment decision in July 2027 and first power from phase one in June 2028^[40]. The Commonwealth referral contains no emissions figure, whereas the Northern Territory's draft terms of reference require the environmental impact statement to report scope 1 and scope 2 emissions by financial year and to identify whether Safeguard obligations apply^[41]. Whether two other large projects would enter the Safeguard Mechanism depends on their grid-connection status, discussed in Annex F.1: Weddell, for which the Northern Territory Government has cited 2,000 MW of gas-fired generating capacity, and Moss Vale, where a 673 MW gas-engine power station is at scoping stage.

How large its emissions could be. Ares could materially affect the calibration. Its estimated gas share ranges from 47% to 78%, depending on the generation and dispatch assumptions. NASA POWER satellite data give a mean annual global horizontal irradiance of 2,271 kWh/m² for the site's grid cell at 17.4°S, 133.4°E (2023 hourly total 2,251)^[39]. At this irradiance, 3,000 MWp of PV would generate approximately 6.48 TWh a year with single-axis tracking or 5.62 TWh with fixed-tilt panels. With annual site demand of 10.5 TWh and 16 GWh of battery storage, the hourly dispatch model gives gas shares of approximately 47% in the tracking case and 53% in the fixed-tilt case. The upper-bound case assumes that the 1,038 MW gas fleet operates at a 90% capacity factor, supplying 78% of annual site demand. Scope 1 emissions are therefore between 2.55 and 4.26 Mt a year. That is 4% to 6% of the 62% point target, and equivalent to approximately 2.1–3.5 years of the annual reserve allowance used for the 2030 settings (1.22 Mt a year). These PV figures are 16–28% below the P50 output of 7,746 GWh a year stated in the proponent's referral report to the Northern Territory Environment Protection Authority^[40], which uses a higher irradiance source; the lower figures are retained here as the more conservative basis, and Annex F.0 explains the difference. Every irradiance, PV, battery and gas assumption is listed in Annex F.0.

How its baseline would be calculated. The baseline follows the new-facility method in Rule s 29 (industry best-practice intensity times production, compilation p 27), and the best-practice intensity for electricity generation is 0.236 t/MWh in Schedule 1 item 57(5) (p 113)^[3]. Three matters depend on regulatory determinations: whether PV output counts, how the facility boundary is drawn, and who the responsible emitter is. Schedule 1 item 57(1)(b) provides that where electricity generation is the only production variable applicable to the facility, only electricity exported from the facility counts, and whether power supplied by a captive plant to a co-located data centre is "exported" depends on the facility boundary determined under the NGER framework. The following baselines are illustrative calculations based on the text of the current Rule. They are not baseline determinations made by the Regulator. If all electricity generated at the site is counted, the estimated FY2034–35 compliance shortfall, covered emissions less the applicable baseline, is between 1.32 and 3.03 Mt. On gas generation alone it is 1.98 to 3.30 Mt. The central case takes fixed-tilt PV, gas at 53% and PV counted, giving a shortfall of 1.69 Mt.

How the calibration could treat those emissions. Once these emissions enter, the calibration can treat them in one of two ways. They could remain within the existing 29.8% share, in which case the other facilities would need to decline by a further 0.20 percentage points per year after 2030. Alternatively, the Safeguard share could be recalculated under footnote 25 of the consultation paper (p 40)^[1], in which case the share becomes 30.2% to 30.4% and the 62% point target rises to 69.7–70.3 Mt. The 29.8% share is based on the average for 2021–22 and 2022–23, before projects of this type were operating at the scale considered here, and the public Commonwealth projection material we reviewed does not explicitly include it. Without a separate adjustment to the Safeguard share, new emissions of this kind fall inside the existing share constraint and raise the decline required of the other facilities.

The part of the reserve relating to large new entrants should therefore disclose its project composition as well as its total. A register by named project would give, for each project, the responsible emitter, the capacity and type of captive fossil generation, the grid-connection determination and its reasons, the expected commissioning year, the owner-declared gas share, and the baseline and covered emissions computed under the Rule. The project-level amounts should reconcile to the total new-entrant component of the reserve. The final decision should also state whether these emissions remain within the existing Safeguard share or lead to a recalculation of that share. The full project analysis, the two tables and the relevant provisions are in Annex F.

6. Ex-post monitoring and recalibration

If the final rate is below the mechanical starting point because of expected on-site abatement, that expectation becomes a substantive component of the calibration. This section sets out a method for measuring the on-site abatement effect each year using only the published register, applies the method to the FY2023–24 and FY2024–25 registers, identifies what can be inferred from the published data and the remaining limitations, and proposes the monitoring and recalibration arrangements that follow.

6.1 What to measure

The annual change in covered emissions can come from facility entry and exit, changes in production and product mix, changes in on-site emissions intensity, and changes in some Rule parameters. To test the on-site abatement assumption used in the calibration, the change at continuing facilities has to be split as far as possible into a scale change and a relative-intensity change. Rule s 11 defines a facility's baseline as the sum over production variables of production times the applicable effective intensity, multiplied by the year's ERC (compilation p 10) ^[3]. Dividing the published baseline by the ERC therefore gives a baseline-implied weighted-output indicator J , and covered $\div J$ gives the facility's emissions intensity I relative to that indicator. By construction, covered emissions equal J times I .

The additive form of the log-mean Divisia index method (LMDI-I) then decomposes the change in covered emissions between two years into a weighted-output effect and a relative-intensity effect. The two effects sum exactly to the actual change with no residual, and facility-level results add up to the panel total (Annex G.1, equations (22)–(25)) ^{[30] [33]}. The weighted-output effect should not be interpreted as a direct measure of physical production: for multi-product facilities it includes changes in product mix, and the annual transition of effective intensity weights under Rule s 13 can move J with production unchanged. The relative-intensity effect is therefore a monitoring indicator obtained from public data and should not be interpreted as a direct estimate of on-site abatement. Three categories of facilities are excluded from the decomposition because J does not provide a consistent weighted-output indicator for them. Annex G.2 explains how this limitation may affect the two decomposition terms. Annex G.4 provides an indicative single-year sensitivity of approximately ± 4 Mt, estimated from measured mine-by-mine production in Queensland and extrapolated from the coal sample to all industries; that figure is not a statistical confidence interval.

6.2 Demonstrating the method on two registers

From FY2023-24 to FY2024-25 the Regulator's covered emissions fell from 135.96 to 132.80 Mt ^{[26] [4]}. Of that 3.16 Mt fall, exits removed 13.22 Mt and entries added 10.80 Mt, and the 184 continuing facilities contributed only 0.44 Mt. The 163 decomposable facilities (95.6% of the continuing base) fell by 0.08 Mt, split into a weighted-output effect of -0.10 Mt and a relative-intensity effect of $+0.02$ Mt (Annex G, Table G.0). An aggregate relative-intensity effect of $+0.02$ Mt is far smaller than the indicative single-year sensitivity of approximately ± 4 Mt, so the current public data cannot determine the direction or size of the aggregate on-site abatement effect.

The industry results are more informative (Table G.1). Relative intensity improved in oil and gas and non-ferrous metals and deteriorated in coal mining and iron and steel. The two largest effects are oil and gas at -0.24 Mt (-0.6% of a base of 42.2 Mt, 25 facilities) and coal mining at $+0.90$ Mt ($+3.5\%$ of a base of 25.5 Mt, 53 facilities). Table G.1 reports the facility counts and both decomposition terms for every industry. For coal mining there is independent evidence outside the register: the intensity of 17 Queensland mines, computed on the state government's measured mine-by-mine production, rose 29% over nine years (Table G.2) ^[29]. Because that estimate uses measured production rather than the baseline-implied indicator J , it is not subject to the same proxy error. The Regulator's published data insights, which also show improvement in oil and gas and deterioration in coal, are consistent with these results ^[27]. The estimated effects for the remaining industries are small relative to the indicative ± 4 Mt aggregate sensitivity. They remain mechanically reproducible from the published data, but their interpretation is subject to the same limitations. This demonstration shows that public data are not yet sufficient to measure the aggregate on-site abatement effect, although the industry-level decomposition can already identify where relative intensity is improving or worsening. The Regulator, which holds facility-level production, could substantially reduce this uncertainty.

6.3 Monitoring and recalibration arrangements

With each annual register, the aggregate and industry-level weighted-output and relative-intensity effects should be published using the identity in section 6.1 and compared with the on-site abatement assumption adopted in the decision document. Performance should be assessed against the abatement pathway adopted when the rate is calibrated. Updating the comparator each year would move the benchmark and weaken the test. The decision document should set out three things in advance: when to reassess if actual cumulative intensity improvement persistently falls short of the calibration assumption; whether a divergence is met by drawing on the reserve, adjusting future rates or revising other assumptions; and the threshold that makes a divergence material. DCCEEW can set that threshold using the facility-level production held by the Regulator and publish it in the final decision. With facility-level production data, the effect of the s 13 transition could be separately identified and removed from the production measure, and the uncertainty would be far below ± 4 Mt. Monitoring can begin with the next annual register, with deviations assessed cumulatively over time.

7. Recommendation

This submission recommends that DCCEEW adopt and publish a transparent, reproducible and periodically reviewable framework for calibrating the post-2030 baseline decline rate. The framework has three consecutive steps.

Step one: fix the starting point. Publish the default rate and the post-2030 TEBA pathway together for both targets. State whether the calibration is based on the FY2034-35 point target or the five-year budget, and publish the inputs and formulas needed for independent reproduction. Under current rules, the paired starting-point rates in this submission are (3.53%, 3.53%) for the 62% target and (5.99%, 5.99%) for the 70% target. This paired result could serve as the starting point, or the Department could recompute it using the same assumptions. Any difference should be traceable to the register version, the s 58B eligible-facility exclusion, the entry distribution, the closure matching or the TEBA path.

Step two: publish the ex-ante adjustments. Publish the quantified effect of each ex-ante adjustment: production, facility entry and exit, TEBA treatment, on-site abatement, below-threshold credits and the reserve. Use these values to complete the calibration bridge in Table 3.1. Large identifiable new entrants should be listed individually. The decision should also state whether their emissions remain within the existing Safeguard share or trigger a recalculation of that share.

Step three: monitor and recalibrate. Publish the aggregate and industry-level relative-intensity effect with each annual register, compare it with the on-site abatement path used in the calibration, and set out in advance the review or recalibration mechanism that applies when the divergence is material.

These measures could be implemented through the review's decision document and accompanying data release, without changing any definition or emissions-intensity value in the current Rule. The algorithm, facility-level tables, project tables and screening tables behind this submission are available to the Department on request for item-by-item reconciliation.

The 3.53% and 5.99% in this submission correspond to step one; the final rate is settled in step two after the adjustments have been quantified and incorporated. Together the three steps form a complete calibration framework: a mechanical starting point, forward-looking adjustments, and verification after implementation.

Part two

Technical annexes

Each annex states the question addressed, identifies the relevant sources and method, and then presents the supporting calculations and results. Quantities marked † are this analysis's arithmetic on published figures.

Annex A	Mechanical calibration using linear percentage-point reductions in the original baseline
Annex B	Reconciliation of the starting-point series with DCCEEW's published series
Annex C	Published estimates disclose their methods to different degrees
Annex D	Effects of facility entry, exit and the minimum baseline on the required rate
Annex E	Inputs not modelled and the parties holding the relevant data
Annex F	Off-grid data centres: project analysis, projects whose Safeguard coverage remains uncertain, and the relevant provisions
Annex G	Relative-intensity decomposition: method, uncertainty and results
Annex H	References and limitations of the evidence base

Terms used in this analysis

Existing facilities / plus new entrants / after confirmed closures / projection	The three population scenarios follow the factors in section 4.2 of the consultation paper: the existing stock only, plus visible new projects, less confirmed closures. All are scenario conventions, not projections
Emissions leaving the covered population / below-threshold credits	Emissions leaving the covered population are the emissions of facilities that fall below the coverage threshold. Below-threshold credits are the SMCs those facilities may earn under s 58B, which depend on the gap between their unfloored baseline and their covered emissions. The two quantities are not equivalent
Linear points / compound	The ERC falls by percentage points of the original baseline each year (0.951 → 0.902 is 4.9 points); a proportional fall on the previous year's value is a different convention with higher numbers
Modelled compliant net / net emissions number	Modelled compliant net emissions are calculated as the smaller of covered emissions and the applicable baseline, assuming full compliance. The statutory net emissions number under NGER Act s 22XK (p 74) also reflects surrenders and multi-year adjustments
Designated large facility / eligible facility	A designated large facility meets the coverage threshold and contributes to the statutory outcomes. An eligible facility under s 58B previously met the threshold but has since fallen below it; it does not contribute to those outcomes
Reserve / calibration result	The reserve is the margin used to absorb uncertainty. The calibration result is the decline rate before the reserve margin is applied; only that pre-reserve result is reported here
ERC (emissions reduction contribution)	The coefficient in the baseline formula that falls each year; an existing facility's baseline = production × intensity determination × ERC (Rule s 11, compilation p 10); the default values are in s 31 (p 28): from FY2023-24, 0.951, 0.902, 0.853, 0.804, 0.755, 0.706, 0.657, then a further 3.285 points a year from 2030-31 ^[3]

TEBA (trade-exposed baseline-adjusted)	A facility the Regulator has found under Rule s 42 (p 36) to be trade-exposed and permitted a slower baseline decline; 21 at present, listed in the register ^[16]
SMC / ACCU	An SMC is a Safeguard Mechanism credit unit, issued to a facility whose emissions are below its baseline and transferable to other facilities for compliance (Rule Part 3A, issuing formula in s 57, pp 45–46); an ACCU is an Australian carbon credit unit, generated by abatement projects and also usable for compliance
BEN (baseline emissions number)	The "baseline emissions number" used in the SMC issuance formula under Rule s 57, defined as the number "if subsection 10(1) had not been made"
Rule s 8 / s 10(1) / s 13	s 8: the coverage threshold, 100,000 t of scope 1 emissions a year (p 9); s 10(1): the minimum baseline, any baseline that computes below 100,000 t is set to 100,000 t (p 10); s 13: the transition proportion, the blend of a facility's own intensity and the industry best-practice intensity moving each year toward best practice (p 12)
Rule s 29 / Schedule 1 item 57	s 29: a new facility's baseline is the industry best-practice intensity times production (p 27); Schedule 1 item 57: the production variable for electricity generation is all electricity produced at the facility, with a best-practice intensity of 0.236 t CO ₂ -e/MWh in subitem (5) (p 113)
Rule s 58B / s 4	s 58B: the definition of an eligible facility, one that once met the threshold, is now below it and can still earn credits (p 48); s 4: the definition of a designated electricity network, listing five networks whose grid-connected generators fall under the sectoral baseline rather than a facility baseline (p 2)
NGER Act s 3(2)(b) / s 22XK	Section 3(2)(b) sets the Safeguard 2030 point target of 100 Mt and the decade budget of 1,233 Mt (printed p 2). Section 22XK defines the net emissions number (p 74) ^[2]

Annex A — Mechanical calibration using linear percentage-point reductions in the original baseline

This annex explains how every number in section 2 is computed. The core facility-level calculation uses the published facility register, the TEBA register and the Safeguard Rule; the production, entry and exit scenarios draw on the additional public sources identified below. The procedure contains no behavioural model, fitted parameter or marginal abatement cost assumption.

A.1 Starting point. The starting population is the 208 facilities reported by the Clean Energy Regulator for FY2024–25^[4]. The register itself has 228 rows. Removing the 19 facilities that are eligible facilities under Rule s 58B (once above the threshold, now below it but still able to earn credits, compilation p 48)^[3] leaves 209 rows. Telfer Gold Mine occupies two of them because of a mid-year change of operator, so the merged count is 208 facilities; covered emissions sum to 132.8 Mt on either count. For each facility, the published baseline emissions number and covered emissions are taken from the FY2024–25 register.

A.2 Advancing the baseline year by year. The Safeguard Rule^[3] defines the baseline as the production of each production variable times its emissions-intensity value times the ERC. Production follows the pathway specified in Annex D.5. Conditional on that pathway, changes in the ERC drive the annual baseline trajectory. The statutory structure of the ERC is a linear fall in percentage points of the original baseline: the published ERC moved from 0.951 in FY2023–24 to 0.902 in FY2024–25, 4.9 percentage points a year, not a compound fall on the previous year's value. The model keeps that structure:

$$ERC_t = 0.902 - 0.049(t - 2025), 2025 \leq t \leq 2030 \quad (1)$$

$$ERC_t = ERC_{2030} - d(t - 2030), t \geq 2031 \quad (2)$$

$$B_t = B_0 \cdot \text{ERC}_t / 0.902 \quad (3)$$

where t is the year in which the financial year ends (FY2024-25 is $t = 2025$; the year index used throughout is $k = t - 2025$), B_0 is the published FY2024-25 baseline, and d is the post-2030 annual decline to be solved for, in percentage points of the original baseline, $d = 0.0353$ being the 3.53% of the main text. Equation (3) is the basic form without production factors or the minimum baseline; the full form is equations (7) and (21). d is the fraction of the original baseline removed each year, not a compound decline relative to the previous year's ERC; every figure in section 2 is on this basis, and converting to a compound annual rate would give higher numbers.

A.3 Minimum baseline. Rule s 10(1) (compilation p 10) ^[3] requires any calculated baseline below 100,000 t to be increased to 100,000 t. The model applies the floor in each year, but only to facilities whose base-year baseline is at least 50,000 t. This modelling simplification prevents facilities already well below the threshold from being mechanically returned to a 100,000 t baseline.

A.4 TEBA facilities. The 21 facilities with a current TEBA determination follow the path in A.9: the register's ERC year by year to expiry, then the default slope from the higher level; all other facilities follow the default path. New TEBA applications, renewals and the minimum decline rate after 2030 are not modelled. Renewal is modelled only as a sensitivity case.

A.5 New entrants. The 62 projects are taken from three Global Energy Monitor trackers ^{[10] [14] [15]}; the selection basis and the itemised list are in Annex D.1. In the sensitivity scenario and the starting point, covered emissions and baselines are estimated from the capacities reported in the trackers, with baselines at the Rule's own best-practice intensities (coal at the 0.00592 t CO₂-e per tonne of run-of-mine coal in Schedule 1 item 17(4), compilation p 80; LNG and iron ore at Schedule 1), a total of 2.59 Mt of baseline a year once all operate, against 17.0 Mt of emissions. Projects enter as follows: the projects accounting for 37% of the emissions have a stated commissioning year after 2025 and enter in that year; those accounting for 16% have a year already past and are treated as awaiting construction, entering in the first modelled year; those accounting for 47% have no year and are spread evenly over ten years. This profile is a modelling convention rather than a project-specific commissioning forecast.

The pipeline enters the model in three steps, written out in equations (4) to (6). Equation (4) gives the share of the pipeline that is operating in each year. Equation (5) gives the emissions and the baseline the pipeline carries once every project is running. Equation (6) multiplies the two: the pipeline's emissions, baseline and net emissions in a given year are the full-build totals times the share operating that year, with the baseline also multiplied by that year's ERC. In the model, covered emissions exceed the applicable baseline for every pipeline project, so pipeline net emissions equal pipeline baselines across the parameter range tested; this follows from the model inputs, not from any statutory identity.

$$s_t = (\sum_{\tau \leq t} E_\tau + E_u \cdot (t - 2025)/10) / E^{\text{pipe}}, s_{2025} = 0, s_{2035} = 1 \quad (4)$$

$$E^{\text{pipe}} = \sum_p Q^{\text{cap}}_p \cdot e_p, B^{\text{pipe}} = \sum_p \hat{Q}_p \cdot \text{EI}^{\text{bp}}_p \quad (5)$$

$$E^{\text{pipe}}_t = E^{\text{pipe}} s_t, B^{\text{pipe}}_t = B^{\text{pipe}} s_t \text{ERC}_t, N^{\text{pipe}}_t = \min \{ E^{\text{pipe}}_t, B^{\text{pipe}}_t \} = B^{\text{pipe}}_t \quad (6)$$

In equation (4), s_t is the share of the pipeline's emissions that is operating in year t . It is built from the commissioning years in the trackers: E_τ is the emissions of the projects whose stated commissioning year is τ (a year already past is treated as 2026), and E_u is the emissions of the projects with no stated year, which are spread evenly over the ten years to 2035. The share is zero in FY2024-25 and reaches one in FY2034-35. In equation (5), Q^{cap}_p is the nameplate capacity of project p , e_p the emissions per unit of capacity for its class of project, $\hat{Q}_p$ the production used to set its baseline, and EI^{bp}_p the Rule's best-practice intensity for that production variable; E^{pipe} and B^{pipe} are the totals over all 62 projects. In equation (6), ERC_t is the year's default ERC (equations (1) and (2)), applied because under Rule s 29 (compilation p 27) ^[3] a new facility's baseline declines with the ERC in the same way as an existing facility's; the pipeline's net emissions are the smaller of its emissions and its baseline, which in this model is always the baseline.

Nameplate capacity is converted into covered emissions, production and baseline estimates as follows for each of the three project classes; the figures are in Annex D.1. For coal (41 projects, 187 Mtpa) the conversion has four steps. First, the operating fleet's ratio of emissions to nameplate capacity is measured: the FY2024-25 register's coal-mining covered emissions of 31.69 Mt divided by the tracker's nameplate capacity of 615 Mtpa for 72 operating mines gives 0.0515 Mt per Mtpa. Second, that ratio is applied to the 187 Mtpa of proposed capacity, giving covered emissions of 9.63 Mt. Third, because the ratio already embeds the operating fleet's utilisation, production is inferred from emissions rather than taken at nameplate: $Q = E \div 0.0653$, where 0.0653 t/t is the transitional intensity that Rule s 17(3) gives coal mines without a facility-specific value (Schedule 1 item 17, compilation p 80). The 41 projects total 147.5 Mt of production, 79% of nameplate, in line with the operating fleet. Fourth, that production multiplied by the best-practice intensity of 0.00592 t/t gives a baseline of 0.87 Mt, against 1.11 Mt if nameplate capacity were used directly.

For LNG (5 trains, 29 Mtpa), nameplate capacity times 54.4 GJ/t and the Schedule 1 default intensity of 0.00414 t/GJ gives emissions of 6.53 Mt; the best-practice intensity of 0.000876 t/GJ gives a baseline of 1.38 Mt.

For iron ore (16 mines, 179 Mtpa), nameplate capacity times the default intensity of 0.00476 t/t gives 0.85 Mt, and the best-practice intensity of 0.00188 t/t gives 0.34 Mt.

Together $E^{\text{pipe}} = 17.0$ Mt and $B^{\text{pipe}} = 2.59$ Mt. The three project classes use different utilisation conventions. Coal emissions use a measured fleet ratio that already embeds utilisation; LNG and iron ore use nameplate capacity with Schedule 1 default intensities and no utilisation discount, which assumes full utilisation and therefore gives an upper-bound estimate of their baselines; recomputed at 90% utilisation, B^{pipe} is about 0.17 Mt lower and the 62% starting point moves by about 0.02 of a point, in the conservative direction. E^{pipe} is far larger than B^{pipe} , so the pipeline's net always equals its baseline: it incurs an annual compliance shortfall from the year of entry and generates no SMCs. No additional production factor is applied to the pipeline, because the pipeline already represents the assumed increase in national output above the base-year level (D.5).

Because the ERC is applied, the pipeline's baseline once all projects operate in FY2034-35 is not 2.59 Mt but $2.59 \times \text{ERC}_{2035}$, about 1.25 Mt at the 62% starting point.

A.6 Compliance assumption and the modelled expression of net. For calibration purposes, and assuming that every facility fully meets its annual compliance obligation, this analysis computes each facility's compliant net as the smaller of its covered emissions and its applicable baseline, then sums. This is a modelling assumption for calibration and does not replace the statutory definition of a facility's net emissions number in NGER Act s 22XK^[2]. This analysis does not model borrowing, multi-year monitoring periods, under-surrender in a year, SMC generation, over-surrender or other inter-period items.

A.7 Target definition and solution method. The target is 2005 national emissions of 607.7 Mt $\times$ (1 – target) $\times$ the Safeguard share of 29.8%, both inputs from the consultation paper^[1], giving 68.8 Mt for the 62% target and 54.3 Mt for the 70% target. Bisection over 0%–30% finds the d at which FY2034-35 compliant net equals the target. The starting point and the three comparison cases moved here from Table 2.3 are in Table A.2.

Formally, facility i 's baseline in year t , the year's compliant net over all facilities, and the calibration condition are

$$B_{i,t}(d) = \max \{ J_{i,t} \cdot \text{ERC}_{i,t}(d), 100\,000 \} \quad (7)$$

$$N_t(d) = \sum_i \min \{ E_{i,t}, B_{i,t}(d) \} + N^{\text{pipe}}_t \quad (8)$$

$$N_{2035}(d^*) = T, T \in \{ 68.8, 54.3 \} \text{ Mt} \quad (9)$$

where $J_{i,t} = J_{i,0} PF_{i,t}$ (equation (21)), $J_{i,0} = B_{i,0} / ERC_{i,0}$, $E_{i,t}$ is covered emissions advanced by the same factor, $ERC_{i,t}(d)$ follows equation (2) or the TEBA path in A.9, and N^{pipe}_t is the pipeline's net (equation (6)). $N_{2035}(d)$ is continuous in d , and the target residual has opposite signs at the ends of the initial interval: at $d = 0$, N is between 87 and 93 Mt, above both targets, and at $d = 0.30$ it is about 20 Mt, below them. Bisection therefore converges, and after 60 steps the interval is sufficiently narrow that further iterations would not affect the reported precision. The min, max and minimum-baseline terms make N piecewise linear and can create kinks or flat segments. Every reported solution was verified to lie on a locally strictly decreasing segment, and the accompanying workbook, which lists equations (7) and (8) facility by facility and contains the solved rates, lets a third party check the residual and reproduce the same solution with Goal Seek. Because N is not differentiable where a facility reaches the floor or its baseline crosses its covered emissions, the cells of Table A.1 cannot be linearly interpolated; intermediate policy cases should be re-solved rather than estimated by interpolating between the reported rates.

Table A.2 — The starting point and three comparison cases, current rules, percentage points a year

Scenario	62% target	70% target
Starting point: production calibrated to OCE commodity volumes, solved to the FY2034-35 point target	3.53	5.99
Reference case: starting point with production flat at FY2024-25	3.90	6.26
Starting point with tracker approval expiry years all treated as closures (10 coal mines)	3.34	5.86
Starting point solved to the five-year budget instead of the point target	2.15	4.59

The 0.37-point gap between the reference case and the starting point comes mainly from thermal coal, LNG, alumina and east-coast gas (D.5). Most tracker years are approval or lease expiries rather than closure decisions (D.2), so they do not enter the starting point. The point-target solution gives five-year net totals of 388 / 344 Mt, below the budgets of 412.8 / 369.3 Mt; the budget solution would overshoot the point target in FY2034-35. The point target is therefore the tighter condition and this analysis calibrates to it.

A.8 Matching closing facilities to the register. The Regulator does not publish stable facility identifiers. This analysis matches by coordinates (obtained through National Pollutant Inventory^[11] site numbers, supplemented from the Geoscience Australia operating-mines dataset^[12]), requires an overlap of substantive name words or mutual nearest-neighbour status as a second independent signal, and checks against production.

Matches are classified as high or medium confidence, and both categories are included. High-confidence matches are those where the coordinates come from an NPI site number, the match distance is ≤ 3 km and the name words overlap, or the match has been manually verified. Medium-confidence matches are those where the coordinates come from the Geoscience Australia dataset by name, the name match is not treated as an independent second signal, and only a distance of ≤ 3 km is accepted. Candidates without a second signal within 10 km are excluded pending manual review. Where one facility corresponds to several mine sites, the facility is not removed because one of them closes.

Matching only attaches the tracker's year to a register facility. Each matched year is then assessed facility by facility to determine whether it represents a confirmed closure, an approval expiry or an approved extension (the three tiers in Annex D.2), and the starting point uses the confirmed tier only. The sensitivity in which tracker years are all adopted is in Annex D.2 and section 1.3.

A.9 TEBA paths and the iso-target curve. A TEBA determination changes only the facility's ERC. The ERC paths of the 21 facilities are taken from the Regulator's TEBA register^[16] year by year to expiry (14 end in FY2025-26 and 7 in FY2026-27; 19 have a reduced rate of 1 point, one of 1.65 and one of 4.21), then follow the default slope, 4.9 points a year before FY2029-30 and d points a year after, from the higher value at expiry, reaching zero in FY2049-50; these are the current rules described in section 3.3.3 of the consultation paper. Other treatments: catch-up pulls the path toward the default from FY2030-31 with a transition proportion $p = (\text{year index} - 2030) \div 20$; reset switches to the default ERC from FY2030-31; renewal extends every determination at its own rate to FY2029-30 and then follows the current path. The iso-target curve (Figure 2.1) represents the TEBA treatment through a single parameter, the post-2030 annual decline d_{TEBA} of the 21 facilities measured from their FY2029-30 ERC under current rules, and for each d_{TEBA} solves for the default rate d_{nonTEBA} that meets the FY2034-35 target; current rules are $d_{\text{TEBA}} = d_{\text{nonTEBA}}$. The slope is about -0.2 (-0.21 for the 62% target, -0.22 for the 70% target): the constrained TEBA facilities' J times production factor totals about 21 Mt and the other constrained facilities about 97 Mt. Points marked on the figure (62% target): current rules (3.53, 3.53); catch-up $\approx$ (4.00, 3.43); reset $\approx$ (5.4, 3.14); TEBA frozen at the FY2029-30 level (0, 4.27); TEBA to zero by 2035 (15.5, 1.41). Facility-level paths are available on request.

Definition and slope of the iso-target curve. Write FY2034-35 compliant net as a function of the two rates, $N(d_T, d_N)$, where d_T is the post-2030 annual decline of the 21 TEBA facilities and d_N the default rate of the other 187. The iso-target curve is the level set of N at the target value, and the slope along the curve and the two partial derivatives are

$$F = \{ (d_T, d_N) : N(d_T, d_N) = T \} \quad (10)$$

$$(dd_N) / (dd_T) = - (\partial N / \partial d_T) / (\partial N / \partial d_N) \quad (11)$$

$$\partial N / \partial d_T \approx -5 \sum_{i \in C_T} J_{i,2035}, \partial N / \partial d_N \approx -5 \sum_{i \in C_N} J_{i,2035} \quad (12)$$

where C_T and C_N are the sets of constrained TEBA and non-TEBA facilities in FY2034-35, constrained meaning net equals baseline and the 100,000 t floor is not reached. N is continuous and non-increasing in both parameters; for each value of d_T , the same bisection procedure identifies the value of d_N for which $N = T$ (50 bisection steps each), and Figure 2.1 joins the solutions for d_T from 0 to 15.5 in steps of 0.5 of a point; if N happens to have a flat segment at the target value the solution is not unique and the reported value is the one the convention selects. Where N is differentiable, equation (11) follows from the implicit function theorem. The approximation in equation (12) comes from the structure: 2030-31 to 2034-35 is five years and d is a decimal, so each extra 0.01 of d (one percentage point) lowers the FY2034-35 baseline of every constrained facility by a further $5 \times 0.01 \times J = 0.05 J$, which as a derivative in d is $-5 \sum J$; unconstrained facilities (covered below baseline) and facilities already at the floor contribute nothing to the derivative. The two constrained sets' J times production factor total about 21 Mt and 97 Mt, a ratio of 0.21 and a slope of -0.21 , consistent with the curve's slope of about -0.21 (62% target) and -0.22 (70% target) obtained from the individual solved points. The two measures used in the main text describe the same underlying trade-off but apply over different ranges. At the current-rules solution, equation (12) gives a local slope of approximately -0.21 and a local default-rate effect of approximately 0.21 percentage points per 1 Mt of TEBA baseline. By contrast, the 0.16 figure in section 2.1 is an average across the larger move from current rules to reset: reset removes approximately 2.41 Mt of TEBA baseline in FY2034-35 and reduces the required default rate by 0.39 percentage points. The two figures differ because the set of constrained facilities changes as the rates move along the iso-target curve.

Table A.1 — Three population scenarios by five TEBA treatments, default rate needed for the 62% / 70% target, percentage points a year

TEBA treatment	Existing facilities: 208, production calibrated to OCE	Plus new entrants: all visible pipeline capacity	Less confirmed closures (starting point)
Current rules: same rate from the higher level after expiry (starting point)	3.46 / 5.96	3.67 / 6.13	3.53 / 5.99
Catch-up: 20-year transition proportion from FY2030-31 (consultation paper option)	3.36 / 5.86	3.57 / 6.03	3.43 / 5.89
Reset: default ERC from FY2030-31	3.06 / 5.56	3.28 / 5.73	3.14 / 5.60

TEBA treatment	Existing facilities: 208, production calibrated to OCE	Plus new entrants: all visible pipeline capacity	Less confirmed closures (starting point)
No concession: all on the default path (counterfactual zero point)	3.06 / 5.56	3.28 / 5.73	3.14 / 5.60
Renewal to FY2029-30 (sensitivity)	3.97 / 6.48	4.17 / 6.64	4.03 / 6.50

"No concession" is the counterfactual zero point; renewal is a sensitivity.

Annex B — Reconciliation of the starting-point series with DCCEE's published series

This annex sets out the arithmetic relationship between this analysis's series and the Department's published series. The reconciliation shows that this analysis's mechanical relationships are compatible with the Department's public series; it does not validate the Department's behavioural assumptions.

B.1 Checking the net basis. Table 9 of *Australia's emissions projections 2025*^[5] reports, for 2030, gross Safeguard emissions of 122 Mt CO₂-e, net unit demand of -26 Mt (the net surrender of ACCUs and SMCs) and net Safeguard emissions of 90 Mt, whereas 122 - 26 = 96 Mt. The 6 Mt difference is SMC generation by facilities below their baselines, reported separately on p 30 of the report. After SMC generation is added back, the published annual series can be reproduced on the modelled compliant-net basis used here. The two series therefore use a consistent compliant-net basis, although they answer different questions. The Department provides a behavioural projection incorporating production, closure and abatement assumptions. This submission instead provides a deliberately narrow mechanical starting point.

The four modelled quantities in Table B.1 and Table 4.1 are all sums of the same facility-level series:

$$N_t = \sum_i \min\{E_{i,t}, B_{i,t}\} + N_t^{\text{pipe}} \quad (13)$$

$$B_t^{\text{agg}} = \sum_i B_{i,t} + B_t^{\text{pipe}} \quad (14)$$

$$R_t = \sum_i E_{i,t} + E_t^{\text{pipe}} \quad (15)$$

$$\Omega = \sum_{t=2021}^{2023} \text{net}_t^{\text{pub}} + \sum_{t=2024}^{2025} \text{net}_t^{\text{reg}} + \sum_{t=2026}^{2030} N_t(0.03285) \quad (16)$$

where N_t is net emissions after compliance, that is equation (8) at a given d ; B_t^{agg} is aggregate baselines; R_t is reference emissions before any on-site abatement; and Ω is the decade budget in NGER Act s 3(2)(b). t is the year in which the financial year ends (2021 means FY2020-21); $\text{net}_t^{\text{pub}}$ is the Department's published figure for the three pre-reform years, $\text{net}_t^{\text{reg}}$ the register's two years, and the remaining years follow the mechanical pathway specified in the starting-point model at the legislated 3.285%. The three sums run over the same population: facilities with confirmed closures leave from their closure year (equation (17)), facilities that fall below the threshold remain (the A.1 convention), and the pipeline enters on the share in equation (4). N_t and B_t^{agg} depend on d ; R_t does not, which is why reference emissions are the quantity that tests the treatment of production and population independently of the rate, and Table B.1 compares them with the Department's.

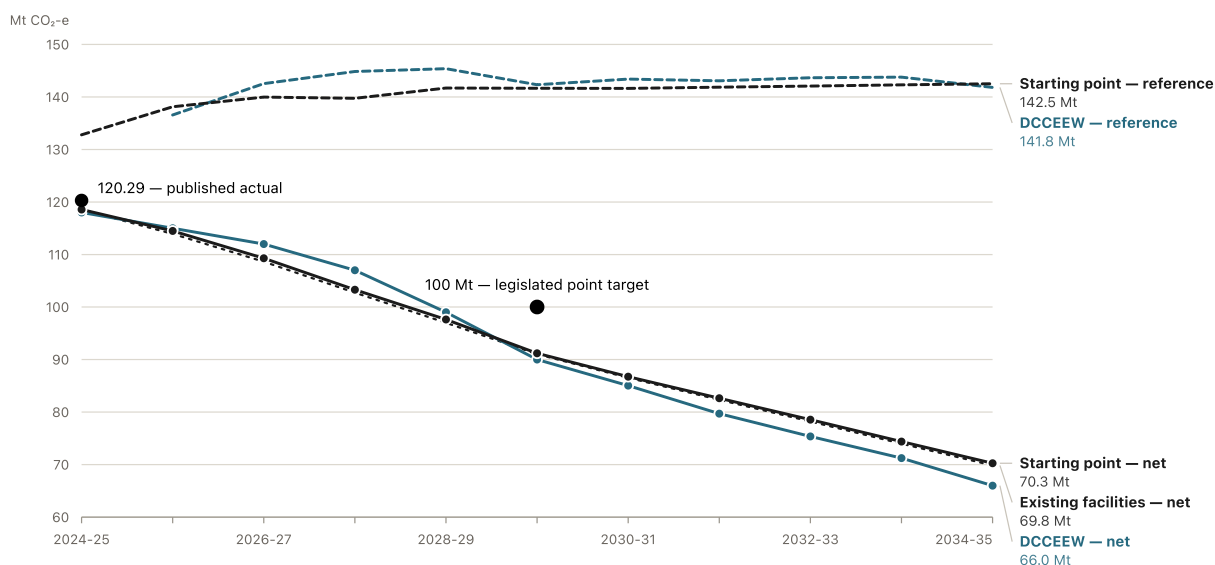
Table B.1 — Comparison of the main series, Mt CO₂-e

Quantity	DCCEE	This analysis, starting point after confirmed closures; TEBA current rules	This analysis, existing facilities	This analysis, plus new entrants
Decade budget, NGER Act s 3(2)(b) net FY2020-21 to FY2029-30, against 1,233 Mt	1,181	1,171.8	1,169.2	1,174.7
Net emissions, FY2029-30 against the legislated 100 Mt point target	90	91.21	91.04	92.32

Quantity	DCCEEW	This analysis, starting point after confirmed closures; TEBA current rules	This analysis, existing facilities	This analysis, plus new entrants
Net emissions, FY2034-35 this analysis at the legislated 3.285%	66	70.25	69.83	71.11
Aggregate baselines, FY2029-30	95.94	93.04	92.87	94.15
Aggregate baselines, FY2034-35 this analysis at the legislated 3.285%	68.99	70.71	70.24	71.52
Reference emissions, FY2029-30 before any on-site abatement; the Department's includes production growth	142.34	141.6	130.8	143.6
Reference emissions, FY2034-35 as above; RepuTex's 2035 BAU is 150	141.81	142.5	127.3	144.3

B.1 addendum: baselines and reference emissions before abatement. Aggregate baselines differ by 2.9 Mt in FY2029-30 and 1.7 Mt in FY2034-35; the FY2034-35 difference includes the additional baseline retained by TEBA facilities under current rules. Reference emissions before any on-site abatement in FY2034-35 are 141.8 Mt for the Department, 142.5 Mt for the starting point and 150 Mt for RepuTex's 2035 BAU. The 0.7 Mt difference indicates close agreement on reference emissions once production is calibrated to OCE national volumes; the flat-production basis gives 147.8 Mt (Annex D.5). The remaining gap between the Department's 66 Mt and the starting point's 70.3 Mt comes mainly from assumptions about on-site abatement, production paths and TEBA treatment. The existing-facilities scenario's reference emissions fall from 132.8 Mt to 127.3 Mt by FY2034-35 through the production factors; the plus-new-entrants scenario reaches 143.6 Mt in FY2029-30 as the pipeline enters on the Annex A.5 profile.

Figure B.1 — Net and reference emissions paths, this analysis and the Department, Mt CO₂-e



Solid lines are net; dashed lines are reference emissions before any on-site abatement. The net paths cross around FY2029-30: before it this analysis is slightly lower (the Department counts unnamed new mines and LNG expansion in 2027-29), after it higher (the Department models abatement, this analysis does not). The two reference-emissions series differ by only 0.7 Mt in FY2034-35: 141.8 Mt for DCCEEW and 142.5 Mt for the starting-point model. Most of the larger difference between their net-emissions paths therefore appears to arise from the Department's assumed abatement pathway. The thin dotted line is the net path of this analysis's existing-facilities scenario.

Under the stated assumptions the decade-budget result differs from the Department's projection by 0.8% (starting point; plus new entrants 0.5%, existing facilities 1.0%); aggregate baselines in FY2029–30 differ by 2.9 Mt. The Department's reference-emissions series was also rebuilt from the published register, the Department's published run-of-mine coal production path^[6] and a near-term production check^[13], with an eleven-year mean difference of 0.9%.

B.2 Where the modelled series is weaker. The modelled series is 1–3 Mt below the Department in FY2026–27 to FY2028–29: the Department counts unnamed new mines and LNG expansion in those years, whereas the pipeline here enters gradually on the Annex A.5 profile. The full-compliance assumption reduces the modelled base-year net figure by approximately 1.7 Mt relative to the reported outcome, because the mechanism allows deferral. The model does not include abatement.

Annex C — Published estimates disclose their methods to different degrees

This annex lists the source-reported values of the estimates on record, the values restated in this submission, the target ranges and the conventions. Heliovolcan's restatements are presented alongside, rather than in place of, the source-reported figures.

Table C.1 — Published estimates of the post-2030 baseline decline rate

Source	62%	70%	Method	Verification supported by public material
This analysis — existing facilities 208 facilities, production calibrated to OCE; TEBA current rules	3.46	5.96	Published register + statutory baseline formula + TEBA register paths; existing facilities only.	Fully reproducible from the cited public inputs
This analysis — plus new entrants +62 projects, baselines at the Rule's own best-practice intensities	3.67	6.13	As above, plus the projects in public trackers; coal at the 0.00592 t/t of Schedule 1 item 17(4).	Fully reproducible from the cited public inputs
This analysis — after confirmed closures (proposed starting point) seven owner-announced or completed closures	3.53	5.99	As above, less owner-announced or completed closures: 3 coal mines and 4 other facilities, 2.03 Mt covered (Annex D.2).	Fully reproducible from the cited public inputs
This analysis — tracker closure years all adopted sensitivity	3.34	5.86	As above, but treating every self-reported tracker closure year as a closure, removing 10 coal mines and 5.88 Mt (matches of high or medium confidence, Annex A.8); on checking, 7 of them are only approval expiry years.	Fully reproducible from the cited public inputs
This analysis — flat production sensitivity	3.90	6.26	As the starting point, but with every facility's production flat at FY2024–25 rather than calibrated to OCE national volumes (Annex D.5).	Fully reproducible from the cited public inputs
RepuTex for ACF and the Climate Council, March 2026, p 17	3.5	6.8	A-EEM behavioural model. The source states it excludes the emissions reserve.	Published result only
EY ^[8] for IGCC, August 2026, pp 10, 23	5	7	Facility-level MACC with uptake decided by NPV. Abatement costs mainly from international estimates. Reserve treatment not disclosed.	Published result only

Source	62%	70%	Method	Verification supported by public material
Australian Energy Producers ^[9] CCA 2025 issues paper, p 2 — restated by this analysis	6.7	9.5	Proportional-share method. The source reports 8–14% for a 65–75% target using a compound-rate convention; 6.7–9.5 is the restatement on the adopted target and a linear basis. Reserve treatment not disclosed.	Partial verification
Climate Change Authority commissioned; delivery November 2026	—	—	Delivered after this consultation closes.	Not yet published
Statutory indicative rate	3.285	3.285	Legislated.	—

Note on the restatement: the Australian Energy Producers row gives the source figure and the restatement side by side; the restatement does not replace the source. Its 8–14 is for a 65–75% target that was not adopted, read on a compound basis, whereas the mechanism's contribution falls **linearly on the original baseline**: 0.951 to 0.902 is 4.9 *points* of the original baseline, not "4.9% less than last year". Restated on the adopted target and the mechanism's own arithmetic it is 6.7–9.5.

Annex D — Effects of facility entry, exit and the minimum baseline on the required rate

This annex documents the sources underlying the facility-population scenarios reported in Table 2.3 and the corresponding columns of Table A.1. It also examines two quantities relevant to the calibration period that are not included in the starting-point calculation.

D.1 New projects: sources and list. All 62 projects come from three public Global Energy Monitor trackers. The Global Coal Mine Tracker, May 2026 edition^[10], contributes the Australian mine sites with status proposed and a capacity figure: 41 sites, 187 Mtpa in total. The Global Gas Infrastructure Tracker — LNG Terminals, September 2025 edition^[14], contributes the Australian export terminals counted by train, excluding import regasification terminals: 5 trains, 29 Mtpa. The Global Iron Ore Mines Tracker, August 2025 edition^[15], contributes the Australian mine sites with status proposed and a capacity figure: 16 sites, 179 Mtpa. Annex A.5, equation (5) and its explanation, describes the conversion from capacity to covered emissions and baseline. In aggregate, the coal projects contribute emissions of 9.63 Mt (from the operating fleet's measured ratio of 0.0515 Mt per Mtpa) and a baseline of 0.87 Mt (from the emissions-implied production of 147.5 Mt, 79% of nameplate, times the best-practice intensity). LNG and iron ore, at nameplate capacity times the Schedule 1 default and best-practice intensities, contribute emissions of 6.53 and 0.85 Mt and baselines of 1.38 and 0.34 Mt. Together the pipeline is 17.0 Mt of emissions and 2.59 Mt of baseline. The trackers list several expansions and continuation projects at existing mines as separate rows (those sharing a name with an existing register facility); all such projects are treated as additions to capacity, which moves the rate in the same direction as production growth at existing facilities. Entry timing is in Annex A.5.

Coal (41; Mtpa, tracker's self-reported commissioning year in brackets): Moranbah South 18.0 (2026), Winchester South 17.0, Carmichael 10.0 and 6.0 (two tracker rows), Kevin's Corner 10.0, Northern Galilee 10.0, Elimatta 8.2 (2027), Caval Ridge 8.0 (2027), Corvus 8.0, West Muswellbrook 7.2, Eagle Downs 7.0, Moolarben 5.5 (2027), Rolleston 5.0, Wallarah 2 5.0 (2027), Mount Pleasant 4.8 (2026), Isaac Plains South 4.5, Glendell 4.0, Meandu 4.0 (2026), Mt Thorley Warkworth 4.0, Newstan 4.0, Vickery 4.0, Millennium 3.6, Wards Well 3.5 (2027), Mangoola 3.5, Hillalong 3.0, Baralaba 2.5 (2028), Cameby Downs 2.4, Vulcan 1.9, Dysart East 1.9, Moorlands 1.9 (2027), Gemini 1.5, Karin 1.4, Springsure 1.2, Star 1.2, Wilpinjong 0.9, Boggabri 0.5 (2029), Isaac River 0.5, Mandalong 0.5, Commodore 0.4, Callan 0.3, Cullenswood 0.1. **LNG (5):** Northern Territory LNG 6.6 (2029), Darwin LNG T2 6.0, Browse FLNG T1 and T2 5.7 each (tracker year 2023, already past), Pluto T2 5.0 (2026, under construction). **Iron ore (16):** Nyidinghu 40.0 (2030), Rhodes Ridge 40.0 (2031), Marillana 20.0 (2025, already past), Mulga Downs 12.0, Central Eyre 12.0, Beebyn-W11 10.0, Lamb Creek 10.0 (2026), McPhee Creek 9.7 (2026), Southdown 5.0 (2026), Razorback 5.0, Roper Valley 4.0, Robe Mesa 3.5 (2025, already past), Ridley 3.0, Hawks Nest 3.0, Yarram 1.2, Speedway 1.0. Projects without a year have no commissioning year in the tracker.

D.2 Facility exits: sources, checks and list. Closure candidates were identified from two sources. For coal, the source is the self-reported closure-year field of the Global Coal Mine Tracker, May 2026 edition^[10]. Of the tracker's 88 operating Australian mine sites, 48 have a closure year recorded in the tracker; of the register's 66 coal-mining facilities (excluding s 58B), 42 match one-to-one to a mine site (Annex A.8), and 10 of those have a closure year of 2035 or earlier. For non-coal facilities, the remaining 143 register facilities were searched individually for company announcements, administrator announcements and regulatory notices (such as NOPSEMA decommissioning environment plans). Each candidate was then checked against three categories of primary evidence: state approval or lease expiry dates; EPBC and state extension applications; and disclosures by the owner. Candidates were sorted into three tiers: **confirmed** (the owner has announced closure or production has ceased), **approval expiry** (the year is only the end of the current approval or lease, with an extension applied for or available), and **extension approved** (the approval already runs beyond 2035). The starting point removes only the "confirmed" tier; a facility with closure year Y leaves from $FY(Y-1)-Y$.

Formally, a facility i with a confirmed closure is removed from the model population in full from the adopted closure year Y_i :

$$E_{i,t} = B_{i,t} = 0, t \geq Y_i \quad (17)$$

where Y_i is the closure year adopted in D.2 (leaving from $FY(Y-1)-Y$, that is $t \geq Y$). Equation (17) means the facility's contribution to the model totals is set to zero, not that its actual emissions or baseline become zero; for $t < Y_i$ it is treated like every other facility. Removal is not proportional, and the facility's emissions are not transferred to other facilities; whether it can earn SMCs under s 58B after falling below the threshold is not modelled. A case like Yarwun, a 40% production cut without closure, is a production item under D.5 and does not follow this equation.

The seven facilities removed (confirmed: announced or completed), by register name and adopted year: Mount Arthur (2030; BHP decision of 2022, NSW consent extended to June 2030), Ashton underground (2028; Yancoal announced a staged closure on 30 June 2026), Myuna (2029; its sole customer, the Eraring power station, retires in April 2029 and the coal supply arrangement ends with it; no formal closure notice was identified; closure is inferred from the announced retirement of its sole customer and the end of the associated supply arrangement), Kwinana alumina refinery (2026; production ceased in 2024, permanent closure announced in September 2025), Liberty Bell Bay (2027; the administrator ordered closure on 16 July 2026), Gove (2030; Rio Tinto: "last shipment of bauxite in 2029"), Ningaloo Vision FPSO (2026; ceased production and departed in August 2025). Covered emissions total 2.03 Mt. The rate needed for the 62% target moves from 3.67% to 3.53% and for the 70% target from 6.13% to 5.99%.

Seven coal mines had tracker closure years but were retained after review: Lake Vermont (2028 is the transition from open cut to underground; the Meadowbrook project adds about 20 years), Metropolitan (approved to 2032; an extension to 2044 was applied for in September 2026), Maules Creek (approved to 2034; an EIS for extension to 2044 was lodged in June 2025), Kestrel (Yancoal, the pending acquirer of an 80% interest, stated a 25-year life-of-mine plan in April 2026), Moranbah North (2033 has no approval or owner basis; longwall production restarted from late 2025 and a sale to Dhilmar is pending), Jax (mining lease expires 2034, no closure statement), Poitrel (about 6 years of reserves, no mine-life statement). Outside the list are Boggabri (2036 → 2040, extension approved), Goonyella-Riverside (2027 has no basis; 448 Mt of reserves) and Hunter Valley Operations (extension to 2042/2045 awaiting IPC determination).

Sensitivity: tracker years all adopted. Removing all 10 of the coal mines above (matches of high or medium confidence, 5.88 Mt) on their tracker years gives a starting point of 3.34% (62%) / 5.86% (70%), 0.19 / 0.13 of a point below the adopted values. The applicable years in this tier change as extension decisions are made, so the results of this sensitivity may change as extension and approval decisions are updated.

Directional effects not included in the starting point. First, Rio Tinto's Yarwun (2.05 Mt) cuts production by 40% from October 2026 and extends its life to 2035. This is a production item rather than an exit; including the production reduction would lower the required rate by approximately 0.08 percentage points. Second, facilities awaiting decisions: the Whyalla steelworks (1.29 Mt, in administration, blast furnace idled), Nyrstar Port Pirie (0.43, government support to end-2026), the Bell Bay aluminium smelter (0.34, power contract expires end-2026), Mount Isa copper smelting (0.33, government support to 2028), the Geelong and Lytton refineries (1.71; their fuel-security commitment end dates are reported as 2027 and 2028 but were not confirmed from primary sources for this submission) and Gippsland/Longford (1.63; the reported timing of Longford plant retirements was likewise not confirmed), about 5.7 Mt in total; several of these individual facilities are comparable in scale to, or larger than, the confirmed coal closures combined. Third, four mine sites with a recorded tracker year (Ulan, Rix's Creek, Chain Valley, Wilpinjong) could not be matched to the register, and 61 candidate matches still require manual review. The uncertainty is therefore two-sided: if approval expiries become closures or pending closure decisions are confirmed, the starting-point rate may be overstated; if confirmed closures are extended, it may be understated.

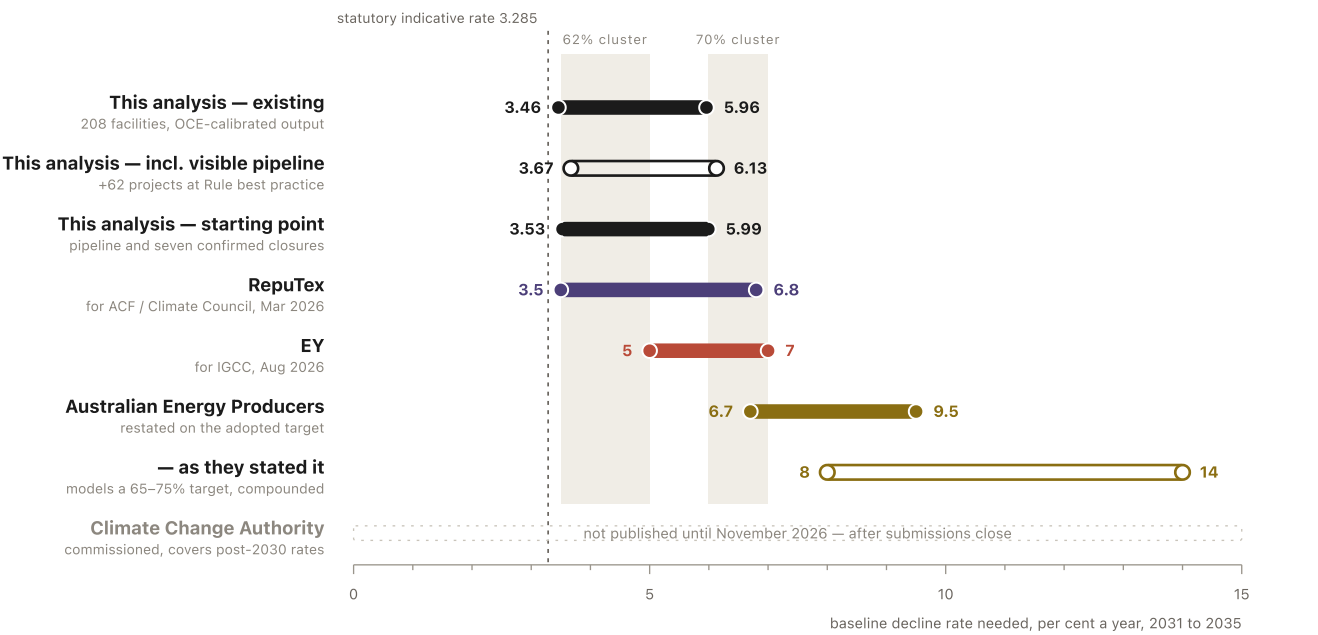
D.3 Emissions that may leave the covered population. Under the NGER Act definitions^[2], the statutory outcomes count designated large facilities only; eligible facilities under s 58B are outside them. Under different abatement assumptions, the emissions leaving the covered population are:

Table D.1 — Emissions leaving the covered population, Mt CO₂-e

Abatement assumption	Basis	FY2034-35	FY2039-40	Share of FY2039-40 net
Zero additional abatement	Starting-point assumption	0.54	0.54	1.1%
1% per year	Sensitivity	1.64	2.15	4.3%
3% per year	DCCEEW projection assumption	3.53	4.71	9.7%
5% per year	Sensitivity	5.13	6.42	14.1%

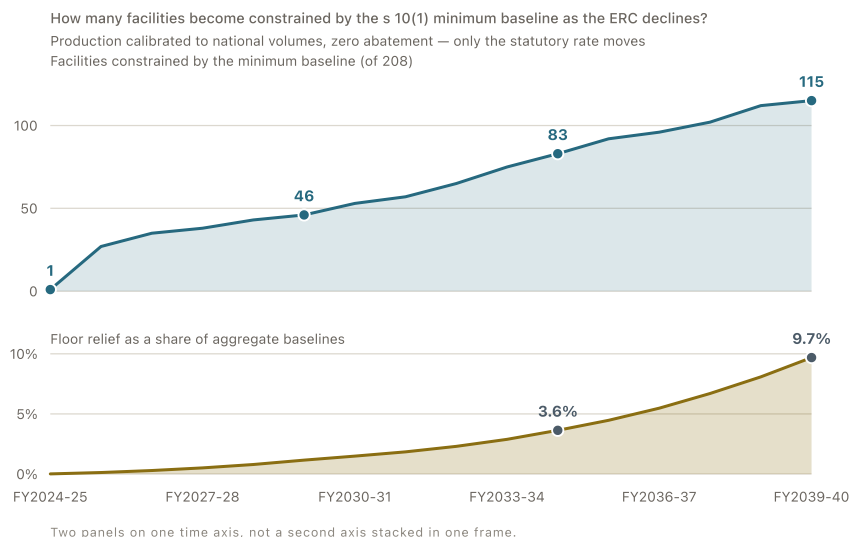
These are emissions leaving the covered population, not the SMCs earned by below-threshold facilities. Those credits depend on the gap between each facility's unfloored baseline and its covered emissions and on the conditions in s 58B, and would need a separate facility-level calculation that this analysis has not made. The calibration period is 2030-31 to 2034-35; the main text cites the FY2034-35 column.

Figure C.1 — Published estimates of the post-2030 baseline decline rate: each bar runs from the rate needed for 62% to the rate for 70%, % a year



Each bar is one estimate, from the 62% target on the left to the 70% target on the right; the shaded bands are the ranges of the comparable estimates at each target; the dashed line is the legislated indicative rate. The estimates differ in population, abatement, reserve, TEBA and linear versus compound conventions, so the positions differ because of assumptions, not targets. Australian Energy Producers' restated figure still falls outside the band, for the reasons in Annex C. The bottom row is the statutory advisory body, whose commissioned modelling is delivered after this consultation closes.

Figure D.1 — Facilities constrained by the s 10(1) minimum baseline, and the resulting baseline relief as a share of aggregate baselines



How many facilities are constrained by the s 10(1) minimum baseline, and the resulting amount of baseline relief.

Production calibrated to national volumes, zero abatement: in this scenario the declining ERC is the only driver of the increase. The two panels share a common time axis but show different measures: facility counts and baseline relief as a percentage of aggregate baselines.

D.4 Minimum baseline. In the existing-facilities scenario (production calibrated to national volumes), the number of facilities constrained by the 100,000 t minimum rises from 1 in the base year to 46 in FY2029-30 and 83 in FY2034-35. The resulting floor relief is equivalent to 3.6% of aggregate baselines in that year; in FY2039-40, beyond this calibration period, it is 115 facilities and 9.7%. For these facilities the decline rate no longer affects the baseline directly. No change to s 10(1) is proposed, only that its effect be quantified in the calibration.

D.5 Production factors: applying OCE national volumes to the existing stock by commodity. Assumption two (Table 2.1 in section 2) is implemented as follows. Each of the 208 register facilities is assigned a commodity key; the detailed mapping is available on request. Coal mines are split between thermal and metallurgical coal by the tracker's Coal Grade and metallurgical share; oil and gas into LNG plants, Queensland CSG upstream, east-coast gas, west-coast gas and oil fields; metals are assigned by facility name; and sectors without a corresponding OCE series, including cement, chemicals, refining and transport, are assigned to "other" and held at 1. For each commodity the Australian national volume is taken from the forecast data table of OCE's *Resources and Energy Quarterly*, June 2026 edition^[18], indexed to FY2024-25 = 1 through FY2030-31; for FY2031-32 to 2034-35 thermal coal follows the IEA WEO 2025 STEPS judgement that Australian thermal coal exports fall about 20% from 2024 to 2035^[19], linearly to 0.80, and all other commodity indices are held constant. The model follows the asymmetric production treatment used in the Department's methodology^[17]: where the national volume is below the base year, existing facilities scale down in proportion; where it is above, the increment goes to "unnamed new mines" and existing facilities are not scaled up, that is, the factor is capped at 1.0. The factor multiplies the facility's output index J , so covered emissions and baseline move together. The same production factors are applied to the weighted-output indicator J for TEBA facilities.

The underlying allocation rule is as follows: when national commodity volume falls, the existing-facility population scales down proportionally; when it rises, existing facilities keep their base-year scale and the increment is assigned to new entrants. Formally, the production factor for commodity c in year t , its assignment to facility i , and how it enters the facility's series:

$$PF_{c,t} = \min\{1, Q_{c,t} / Q_{c,2025}\}, 2025 \leq t \leq 2031 \quad (18)$$

$$PF_{th,t} = PF_{th,2031} + (0.80 - PF_{th,2031}) (t - 2031)/4, 2032 \leq t \leq 2035; \text{ other commodities } PF_{c,t} = PF_{c,2031} \quad (19)$$

$$PF_{i,t} = m_i PF_{met,t} + (1 - m_i) PF_{th,t} \text{ (coal mines); } PF_{i,t} = PF_{c(i),t} \text{ (others)} \quad (20)$$

$$J_{i,t} = J_{i,0} PF_{i,t}, E_{i,t} = E_{i,0} PF_{i,t}, B_{i,t} = \max \{ J_{i,t} \cdot ERC_{i,t}, 100\,000 \} \quad (21)$$

where $Q_{c,t}$ is OCE's national volume forecast for commodity c ^[18], th and met are thermal and metallurgical coal, m_i is coal mine i 's metallurgical share, $c(i)$ is the commodity facility i is assigned to, facilities assigned to "other" have $PF \equiv 1$, and 0.80 is the IEA's judgement for Australian thermal coal exports in 2035^[19]. The cap $\min\{1, \cdot\}$ in equation (18) is the asymmetric rule of the Department's methodology^[17]: below the base year, the stock scales down in proportion; above it, the increment goes to new entrants and the stock is unchanged. Equation (21) multiplies J and E by the same factor, so the production factor changes a facility's scale but not its intensity relative to its own benchmark, consistent with the definition of the relative-intensity effect in section 6. The IEA judgement for metallurgical coal (about +5% by 2035) has no effect once capped. The uncapped convention (increments added to the stock) and the flat-production convention are sensitivities, in Tables A.2 and D.2.

Table D.2 — Production factors: national volume index by commodity (FY2024-25 = 1) and the capped values applied to the existing stock

Commodity (OCE series)	FY2030-31 index	Capped, FY2030-31	Capped, FY2034-35
Thermal coal (export volume)	0.964	0.964	0.800
Metallurgical coal (export volume)	1.070	1.000	1.000
LNG (export volume)	0.969	0.969	0.969
East-coast gas (production)	0.890	0.890	0.890
West-coast gas (production)	0.944	0.944	0.944
Crude oil and condensate (production)	0.647	0.647	0.647
Alumina (production)	0.956	0.956	0.956
Primary aluminium (production)	1.062	1.000	1.000
Bauxite (production)	1.059	1.000	1.000
Iron ore (production)	1.006	1.000	1.000
Crude steel (production)	1.231	1.000	1.000
Gold (mine production)	1.285	1.000	1.000
Copper (mine production)	1.355	1.000	1.000
Zinc and lead (zinc mine production)	0.922	0.922	0.922
Nickel (refined production)	0.831	0.831	0.831
Lithium (mine production)	1.599	1.000	1.000

†. Index FY2024-25 = 1. Capped = $\min(1, \text{index})$. Mine-site power stations follow their host commodity; mineral sands, manganese and "other" have no OCE series and are fixed at 1.

Calibrating production to national commodity volumes lowers the required rates (TEBA current rules, 62% / 70%) for the existing-facility scenario from 3.85 / 6.24 to 3.46 / 5.96. For the pipeline scenario the corresponding rates fall from 4.04 / 6.39 to 3.67 / 6.13. The starting-point rates fall from 3.90 / 6.26 to **3.53 / 5.99** (adopted). Uncapped, with the existing-facility population scaled up with national volume, the starting point is 3.82 / 6.19. Run to FY2034-35 at 3.285%, the starting point's net moves from 72.64 Mt to 70.25 Mt, and reference emissions before abatement from 147.8 Mt to 142.5 Mt, only 0.7 Mt above DCCEEW's estimate of 141.8 Mt (Table B.1). Two limitations should be noted. OCE's volumes run to 2030-31, and after 2031 only thermal coal has a citable external judgement; national commodity projections already incorporate some expected closures and new capacity, so applying them to the stock overlaps slightly with the closures and pipeline listed separately here.

A question of statutory interpretation for DCCEEW to confirm: the formula for issuing SMCs in Rule s 57 (compilation pp 45–46) defines BEN as the baseline "if subsection 10(1) had not been made". On this analysis's reading, the minimum baseline relieves a facility's obligation but does not generate SMCs for it.

Annex E — Inputs not modelled and the parties holding the relevant data

This annex consolidates the inputs discussed in sections 2 to 4 that cannot be established from the public record, states how each is treated here and who currently holds it. Recommendation 1 proposes that the final decision document disclose how each of these inputs is treated.

Table E.1 — Inputs needed for the calibration that the public record cannot establish

Input	Treatment in the model	Data holder
Post-2030 production at existing facilities	Calibrated to OCE's public national commodity volumes, stock scaled down but not up (Annex D.5); flat production as a sensitivity	Each responsible emitter; the Regulator holds historical production but does not publish it; the Department uses OCE mine-by-mine figures and Wood Mackenzie, neither public
Commissioning probability and timing of new projects	Sensitivity scenario with all projects entering; timing in the three-part treatment of Annex A.5	Project owners; DCCEEW's projections carry their own assumptions
Closures of non-coal facilities	The four confirmed (announced or completed) closures are included; about 5.7 Mt of pending decisions and the Yarwun cut are not	Each responsible emitter
TEBA applications, renewals and the minimum decline rate after 2030	Not modelled; base-year TEBA is embedded in the starting point	DCCEEW / the Regulator (application data)
Policy changes from this review	Not modelled	DCCEEW
SMCs issued to below-threshold facilities	Not quantified; only the emissions leaving the population are quantified	The Regulator (facility-level unfloored baselines)
Size and composition of the reserve	Excluded	DCCEEW
TEBA claw-back treatment (catch-up / reset / other)	Starting point under current rules; other treatments in Table 2.2 and Figure 2.1	This review (sections 3.3.3–3.3.4)
Annual transition of the s 13 intensity blend	Frozen at FY2024-25; not quantified	The Regulator (default and facility-specific values and production by production variable)
Expected TEBA uptake used in the decline-rate calibration	Not available in the public record; only the observed concession is reported	DCCEEW
Corporate behaviour, technology adoption and marginal abatement cost	Not modelled	Each responsible emitter; each published estimate carries its own assumptions

"Not modelled" and "not quantified" in the table are modelling choices, not judgements that the input does not exist or does not matter.

Table E.2 — Six assumptions: value adopted, basis, and the effect of alternative assumptions

Assumption	Value in this analysis	Basis	Effect on the required rate under alternative assumptions
Target	Both the 62% and 70% targets; share 29.8%; 2005 emissions 607.7 Mt; the compliance condition is the FY2034-35 point target (net = 68.8 / 54.3 Mt) as a conservative calibration convention ; the consultation paper also gives a five-year budget (412.8 / 369.3 Mt), and the budget-basis solution is in Annex A, Table A.2	Consultation paper ^[1] section 4.1	Fixed once the target is set; no preferred point within the target range is adopted. Solving to the budget is 1.4 points lower, and the point-target solution's five-year total of 388 Mt is already within the budget, so the point target is the tighter condition; which of the two is the legal compliance condition is for the Department to state
1 · Covered population	209 register rows (228 rows less 19 eligible facilities under s 58B; the Regulator counts 208 facilities, the difference being Telfer Gold Mine on two rows after a mid-year change of operator, with covered emissions of 132.8 Mt on either count), plus all 62 publicly visible pipeline projects without probability weighting on a stated entry distribution, less seven owner-announced or completed closures (3 coal mines and 4 other facilities, 2.03 Mt covered; Annex D.2)	Register ^[4] ; three GEM trackers ^{[10] [14] [15]}	Pipeline +0.21 points (62%) / +0.16 (70%); confirmed closures -0.14 / -0.14 ⁺ ; treating tracker approval expiry years as closures (10 coal mines, 5.88 Mt) a further -0.19 / -0.13 (sensitivity, Annex D.2); 6 pending facilities (5.7 Mt) not counted
2 · Production	Each facility's output index <i>J</i> follows the national volume for its commodity: OCE volumes to 2030-31, after 2031 thermal coal per the IEA and the rest flat; the part of national volume above the base year goes to new entrants and the stock is not scaled up (factor capped at 1.0)	OCE REQ ^[18] , IEA WEO ^[19] , the calibration rule in the Department's methodology ^[17] (Annex D.5)	Flat production (reference case) +0.37 / +0.27 points; stock scaled up with national volume (uncapped) +0.29 / +0.20; mine-by-mine projections: held by the Department, not public (section 3.4)
3 · Baseline formula	The ERC falls linearly in percentage points of the original baseline; the s 10(1) minimum baseline of 100,000 t; each facility's intensity blend frozen at FY2024-25	Rule ^[3] ss 10(1), 11(1), 13	Legislated. The s 13 transition (standard 20:80 → 100:0, coal 10:90 → 50:50) may differ materially in its effects across facilities even if the aggregate effect is small; not quantified by this analysis (section 3.4)
4 · Compliance	Full compliance, net = min(covered, baseline); no abatement, borrowing or inter-period items modelled	Modelling assumption (Annex A.6)	Actual on-site abatement lowers it; section 6 gives the current measurement: the aggregate relative-intensity effect of +0.02 Mt is far smaller than the indicative single-year sensitivity of approximately ±4 Mt extrapolated from the coal sample, so the current public data cannot determine the direction or size of the aggregate on-site abatement effect
5 · Items excluded	No reserve, no new TEBA applications or renewals after 2030, no below-threshold credits	Factors three, five and six in section 4.2	The reserve raises it; renewal raises the default rate to 4.03 (sensitivity); below-threshold credits need a separate facility-level calculation

Assumption	Value in this analysis	Basis	Effect on the required rate under alternative assumptions
6 · TEBA treatment	The 21 current determinations follow the register ERC to expiry (14 end in FY2025–26, 7 in FY2026–27), then decline at the default slope from the higher level, at the same rate as non-TEBA facilities after 2030	TEBA register ^[16] ; consultation paper section 3.3.3	Catch-up (20 years from 2031) 3.43; reset 3.14; iso-target curve in Figure 2.1; moving from current rules to reset lowers the default rate by an average of 0.16 of a point per Mt of TEBA baseline removed [†]

Annex F — Off-grid data centres: project analysis, projects whose Safeguard coverage remains uncertain, and the relevant provisions

F.0 The size of Ares and the two treatments of its emissions in the calibration. The two tables of section 5 in the main text.

Values and sources: site irradiance, PV output and gas share. The material reviewed for this submission does not state how much of the electricity gas will supply; the share is estimated here from site data, and every value below is an assumption of this analysis, not a proponent figure. Irradiance is taken from NASA POWER satellite data^[39] (SYN1deg grid, about 1°, approximate site coordinates 17.4°S, 133.4°E, Murranji Station, Barkly): the 2001–2020 mean annual global horizontal irradiance is 2,271 kWh/m² and the 2023 hourly total 2,251 kWh/m², within 1% of each other; the hourly energy balance uses 2023. PV specific yield uses a performance ratio of 0.80, a tilt gain relative to the horizontal of 1.04 for fixed tilt and 1.20 for single-axis tracking, module temperature equal to air temperature plus 25 °C times the irradiance fraction, and a power temperature coefficient of –0.4%/°C, giving about 1,873 kWh/kWp fixed-tilt and 2,161 kWh/kWp tracking, so 3,000 MWp generates 5.62–6.48 TWh a year. The proponent's referral report to the Northern Territory Environment Protection Authority^[40] states a combined P50 output of about 7,746 GWh a year for the full build (about 2,580 kWh/kWp) on a Solargis site irradiance of 2,300–2,400 kWh/m², 16–28% above the figures used here; the difference comes from the irradiance source and the loss assumptions, and the lower figures are retained because they are the more conservative basis for the gas share. The same report cites achievable single-axis capacity factors of 31–34%, whereas its P50 figure implies about 29.5%.

Load is 1,000 MW of IT times a PUE of 1.2, flat all year at 10.5 TWh; the battery is 16 GWh with 2,000 MW of charge and discharge power and a round-trip efficiency of 88%; dispatch order is PV direct to load, surplus to the battery, and shortfall from the battery first and gas second; gas intensity is 0.52 t CO₂-e/MWh (gas turbine).

Hourly results: gas supplies 47% of the load with tracking and 53% fixed-tilt; neither case curtails PV, and doubling the assumed battery capacity does not materially change the modelled gas share, indicating that the result is driven mainly by solar yield rather than by the assumed battery capacity; the simple energy-balance floor (annual PV output divided by load) is 38%–47%, within 10 points of the hourly result. The upper end, 78%, is the 1,038 MW of turbines at a 90% capacity factor.

Phase one is 500 MW of IT, a load of 5.3 TWh a year, served by the same 3,000 MWp of fixed-tilt PV and the same 16 GWh battery, giving a gas share of 11%; the much lower share reflects the full solar and storage capacity being available against half the load, and it should not be read as a scaled version of the full-build case. The stated generation and load capacities are not fully aligned: 1,038 MW of gas capacity is below the 1,200 MW total site load at a PUE of 1.2. If gas is capped at 1,038 MW, 0.57–0.65 TWh of load cannot be met in the hourly dispatch, which suggests the proponent's gas capacity is sized to the 1,000 MW IT load. The model assumes that gas generation supplies the full residual load after solar and battery dispatch, which slightly overstates scope 1 emissions and is therefore conservative for the purpose of this screening analysis.

Table F.1 — The size of Ares under four cases

Case	Gas, TWh	Scope 1, Mt a year	Share of 62% point target	Years of reserve at the 2030 convention	2034-35 baseline, Mt	2034-35 shortfall, Mt
Full load, gas 47% (site irradiance, single-axis tracking, hourly dispatch; lower end)	4.9	2.55	3.7%	2.1 years	1.22	+1.32
Full load, gas 53% (site irradiance, fixed tilt, hourly dispatch; central)	5.6	2.91	4.2%	2.4 years	1.22	+1.69
Full load, gas 78% (1,038 MW of turbines at a 90% capacity factor; upper end)	8.2	4.26	6.2%	3.5 years	1.22	+3.03
Phase one, 500 MW IT, gas 11% (site irradiance, fixed tilt)	0.6	0.31	0.4%	0.3 years	0.61	-0.30

The point target at 62% is 68.8 Mt. The 2030 reserve convention was "emissions linked to production may grow by up to 0.9% a year", which on 2023-24 aggregate baselines of 136 Mt is 1.22 Mt a year. The 2034-35 ERC is 0.493. The shortfall is scope 1 minus baseline; a negative value means emissions below baseline. The proponent has not published how much of the electricity gas will supply; the three cases come from the hourly energy balance on site irradiance and the turbine capacity cap (see the values and sources above). If the baseline counts gas generation only, the three full-load shortfalls are 1.98, 2.26 and 3.30 Mt (Annex F.2).

Table F.2 — Two treatments of Ares's emissions in the calibration once they enter

Treatment	Arithmetic	Implication for emissions outside the project
Treatment 1 — emissions remain within the 29.8% share, treated as Safeguard production growth	The point target of 68.8 Mt is unchanged and Ares's net emissions must be accommodated within it. Because its covered emissions exceed its baseline, net equals baseline, so the three full-load cases give the same result: the other facilities must decline by a further 0.20 of a point a year after 2030 (62% target), or 0.05 of a point for phase one. Computed with the section 2 solver by adding one facility to the starting-point population and re-solving; the starting point itself excludes Ares	The existing 208 facilities
Treatment 2 — the Safeguard share is recalculated under footnote 25	The share is reference-period Safeguard emissions divided by national emissions. Adding Ares to numerator and denominator, the three cases give 30.2%, 30.2% and 30.4%, the point target at 62% rises to 69.7–70.3 Mt, an extra 0.9–1.5 Mt	The corresponding emissions-reduction requirement shifts outside the Safeguard sector, because the national target remains unchanged

The share recomputation uses the reference-period figures in Table 2.1 of the consultation paper, about 137 Mt in 2021-22 and 138.6 Mt in 2022-23, mean 137.9 Mt; the national figure is backed out from 29.8%, giving 462.8 Mt. This is illustrative arithmetic; a recomputation by the Department from inventory values would differ slightly.

F.1 Projects whose entry into the Safeguard Mechanism is uncertain. Weddell is proposed by Beetaloo Digital Pty Ltd, a subsidiary of Beetaloo Energy, which describes its own role as gas and land supplier^[42]; the project is at concept stage. The Northern Territory Government has cited 2,000 MW of gas-fired generating capacity^[24]; the proponent's material reviewed for this submission does not state an IT load or a renewable share. The plant would connect to the Darwin–Katherine network and sell into it. Depending on the assumed capacity factor (40% to 90%) at a gas-turbine intensity of 0.52 t CO₂-e/MWh, 2,000 MW of gas-fired capacity could produce approximately 3.64–8.20 Mt of scope 1 emissions a year; these are screening estimates rather than a project forecast. Under Rule s 4 as it stands, a power station connected to a designated electricity network at any time in a financial year is a grid-connected generator whose emissions fall under the electricity sector's sectoral baseline rather than being covered emissions; that line is 198 Mt, and grid-connected generators emitted only 133.7 Mt in 2024–25^[23]. If it is found to be a grid-connected electricity generator, its emissions would fall under the sectoral baseline and would not create a facility-level baseline obligation; the actual obligation depends on how the sectoral baseline applies in the year. Moss Vale's power station, 673 MW of gas reciprocating engines at scoping stage, is proposed to serve the campus itself; the campus sits inside the Endeavour Energy 11 kV network, and on the same screening basis 673 MW would produce 1.23–2.77 Mt a year. Two items in the planning record bear on the grid-connection question. The proponent's air-quality consultant reports scope 1 emissions of about 112,000 t CO₂-e a year for the first two stages alone, 30 MW of continuous generation, which on its own would exceed the coverage threshold^[43]. The network operator has advised that cabling connecting the gas generators and battery to the grid-connected site "may be considered to be grid connected (ie. embedded generation), unless a break mechanism is provided", and the material reviewed does not state whether such a mechanism is proposed^[44]. The facility boundary and the environmental impact statement are not yet settled, so no conclusion is drawn here. If either is determined to be a facility, its size is comparable to or larger than Ares. No prediction is made about the determinations. Large projects whose power sources are not stated in the public material reviewed here include the 2.4 GW Plumpton in Victoria, Meridien in the Kimberley, Fortescue's 2 GW concept at Port Hedland, the 800 MW Cooke Plains in South Australia and six confidential proposals in Tasmania; the Northern Territory government says it has 12 proponents^[24], of which only Weddell and Ares are named publicly. The register screening method and site-by-site results are available on request.

F.2 Relevant provisions, grouped by decision-maker. Two provisions are in the Safeguard Rule but not discussed in this consultation.

The first is the definition of grid connection: Rule s 4 lists five designated electricity networks, the Darwin to Katherine network among them, and the definition of a grid-connected electricity generator is a single clause, "connected at any time in the financial year"^[3], with no minimum export share and no test of whom the plant mainly serves; the consultation paper does not mention grid-connected, sectoral baseline or designated network anywhere, so no recommendation is made on it because it falls outside the scope of this review; any future register classification of these projects would depend on that grid-connection test.

The second is the baseline for new facilities: s 29 uses best-practice intensity, 0.236 t/MWh for electricity generation under item 57(5), and the default of 0.539 applies only to historical variables at existing facilities; the paper says intensity values are not being revisited this time.

Four matters sit outside the Safeguard Mechanism or are shared among several parties. First, whether the PV output of a hybrid station counts in the item 57(1) production variable; this analysis reads the clause "all electricity produced at the facility" literally as counting it, but item 57(1)(b) provides that where electricity generation is the only production variable applicable to the facility, the electricity counted must be exported from the facility, and whether power supplied by a captive plant to a co-located data centre is "exported" depends on the facility boundary determined under the NGER framework; if the Regulator determines that only gas counts, Ares's three full-load 2034–35 shortfalls become 1.98, 2.26 and 3.30 Mt, and phase one becomes +0.24 Mt. Second, whether a third-party power station inside the fence serving a single load is one facility or two with that load, and who the responsible emitter is, is determined under NGER Act ss 9 and 11 and the Regulator's facility guidance^{[2] [25]}, which indicates that the outcome depends partly on the contractual and operational arrangements. Third, whether multiple units at one site serving one load must be aggregated falls under the anti-avoidance provision in NGER Act s 54B. Fourth, a new production variable for computing load would require the Minister's rule-amendment process, which this review has not listed.

These four determine how the register's baseline, responsible emitter and row count are filled in; this analysis makes no recommendation on any of them.

Annex G — Relative-intensity decomposition: method, exclusions, uncertainty and results

G.0 Reconciliation of the reported change in covered emissions. This section reconciles the figures reported in section 6.2 of the main text.

Table G.0 — Reconciliation of the reported change in covered emissions, FY2023-24 → FY2024-25, Mt CO₂-e

Step	Facilities	ΔE	Note
Regulator's basis: FY2023-24 135.96 → FY2024-25 132.80	219 → 208	−3.16	s 58B excluded; the Regulator's headline is −2.3% ^[27]
Exits (on the FY2023-24 register only)	32	−13.22	This category includes closures, changes of operator and facilities falling below the coverage threshold; they are not distinguished here
Entries (on the FY2024-25 register only)	21	+10.80	Net coverage change −2.43
Name-format changes only	3 pairs	−0.29	Treated as the same facility
Continuing under the same name	184	−0.44	Base 121.26 Mt
Of which excluded (three classes)	21	−0.36	Floor 17, new determination 2, borrowing 2
Decomposable	163	−0.08	Base 115.96 Mt, 95.6% of continuing facilities
Weighted-output effect		−0.10	<i>J</i> down 0.08%
Relative-intensity effect		+0.02	+0.01%; facility median −0.4%, quartiles −7.8% to +5.2%

The two effects are LMDI decomposition results; the "weighted-output effect" is not directly observed physical production, and the limits of interpretation are in G.2 and G.4. The data are only the Regulator's published FY2023-24 and FY2024-25 registers^{[26] [4]} and the intensity determinations table^[28]. Of the 163 facilities, 102 have two or more production variables. Without the three exclusions the relative-intensity effect would be +0.46 Mt, mostly an artefact of floor facilities whose baselines are decoupled from production.

G.1 The decomposition identity. For facility *i*, the relationship in Rule s 11(1) (compilation p 10)^[3] can be written

$$B_{i,t} = \text{ERC}_{i,t} \sum_k Q_{i,k,t} \cdot \text{EI}_{i,k,t} \quad (22)$$

$$J_{i,t} \equiv B_{i,t} / \text{ERC}_{i,t}, I_{i,t} \equiv E_{i,t} / J_{i,t}, E_{i,t} = J_{i,t} I_{i,t} \quad (23)$$

$$\Delta E_i = L(E_{i,t}, E_{i,t-1}) \ln(J_{i,t} / J_{i,t-1}) + L(E_{i,t}, E_{i,t-1}) \ln(I_{i,t} / I_{i,t-1}) \quad (24)$$

$$L(a, b) = (a - b) / \ln(a / b) \ (a \neq b); L(a, a) = a \quad (25)$$

where *B* is the baseline, *E* covered emissions, and *Q_{i,k,t}* and *EI_{i,k,t}* the production and the applicable effective intensity of each production variable *k*; the decomposition is between adjacent years *t* − 1 and *t*, and in this annex *t* = 2025 (FY2024-25) and *t* − 1 = 2024 (FY2023-24). The first term of equation (24) is called the weighted-output effect and the second the relative-intensity effect; their sum equals the facility's change in covered emissions exactly, and the contributions of the 163 facilities add exactly to the panel's change^[30]. Baseline, ERC and covered emissions are published facility by facility and year by year, so *J* and *I* can be computed for every facility without knowing any production figure.

Dividing by the ERC removes the mechanical effect of the baseline decline rate itself (the TEBA concession is already in each facility's published ERC), but it does not remove the effect of every change in the Rule's parameters. If the effective intensity weights of each production variable were constant, J could be read as a fixed-weight output indicator, equivalent in structure to a Laspeyres quantity index^[38] with the Regulator's determined EIs as weights instead of prices; between the two years in this analysis, the s 13 transition may have changed the effective weights of some facilities. J should therefore be interpreted as a baseline-implied weighted-output indicator rather than as directly observed physical production; the "weighted-output effect" is not a pure physical-production effect, and the "relative-intensity effect" should not be interpreted as a direct estimate of on-site abatement. Section G.2 explains how changing intensity weights may affect the two decomposition terms, and section G.4 provides an indicative estimate of the resulting sensitivity.

Equation (24) is the additive form of the log-mean Divisia index method LMDI-I of Ang and co-authors^{[32] [33]}: $E = J \times I$ has the structure of the Kaya identity^[35], and each factor's contribution is the log-mean weight times the logarithmic change in that factor. LMDI-I is used because it is complete, leaving no residual, and facility-level results can be added directly^{[33] [34]}; the log mean L (equation (25)) follows Törnqvist, Vartia and Vartia^[36]. Implementation follows Ang (2015)^[37]: all values are positive, so no zero-value treatment is needed, and results are reported in Mt.

G.2 The transition weights of Rule s 13. Rule s 13 (compilation p 12) provides that for a production variable with a facility-specific intensity value, EI is h times the default value plus $(1 - h)$ times the facility-specific value, with h equal to 0.1 in FY2023-24, 0.2 in FY2024-25 and 1 from 1 July 2029; for coal mines the blend is 5:95 to 10:90 per Table 3.1 of the consultation paper^[1]. The annual change in J is therefore not entirely production, and both LMDI terms are affected by the change in effective intensity weights. The weight change affects the two terms in opposite directions and does not affect their sum, which still equals the actual change in emissions. Because production by production variable and the applicable weights are not fully public, the effect cannot be removed facility by facility; its possible magnitude is instead estimated in G.4 from the sample of coal mines with measured production.

G.3 Three categories excluded. The first comprises 17 facilities whose baselines are constrained by the s 10(1) minimum of 100,000 t, so that baseline and production are decoupled. The second comprises two facilities with new intensity determinations taking effect in FY2024-25, which changed their weights. The third comprises two facilities with borrowing adjustments that shift baseline amounts between years. The three categories total 21 facilities, 5.3 Mt of base emissions and ΔE of -0.36 Mt. Exits and entries are identified by comparing facility names across the two registers, with 3 pairs differing only in name format treated as the same facility; the Regulator's basis excludes eligible facilities under s 58B.

G.4 Estimating the uncertainty magnitude from measured production. The Queensland Department of Resources publishes mine-by-mine production by year^[29]. Gross Raw Output data for the 22 mines with production in both years and no restart or shutdown were matched to the corresponding register facilities, and the production change implied by J was then compared with the measured change: the median absolute gap is 2.2 percentage points (1.8 points for the 14 single-production-variable mines, 2.5 for the 8 multi-variable mines), and 14 of the 22 lie within ± 5 points. Replacing J with measured production changed the estimated relative-intensity effect of these 22 mines (base 9.76 Mt) by 0.34 Mt in total, 3.4% of their base. Applying that proportion to the 163-facility base of 115.96 Mt gives an indicative single-year sensitivity of approximately ± 4.0 Mt. This estimate should be interpreted as a sensitivity range. It is neither a measurement-error estimate nor a statistical confidence interval. In the absence of comparable production data for other industries, the same proportional sensitivity is applied to the broader sample; this extrapolation is a material limitation.

Table G.1 — Weighted-output and relative-intensity effects of the 163 facilities by industry, FY2023-24 → FY2024-25

Industry (ANZSIC)	Facilities	Base Mt	Weighted-output effect, kt	Relative-intensity effect, kt	Relative-intensity effect ÷ base
Oil and gas extraction	25	42.19	-752	-239	-0.57%
Coal mining	53	25.54	-131	+900	+3.53%
Basic non-ferrous metal manufacturing	9	13.64	+98	-226	-1.65%

Industry (ANZSIC)	Facilities	Base Mt	Weighted-output effect, kt	Relative-intensity effect, kt	Relative-intensity effect ÷ base
Metal ore mining	32	8.37	+111	+113	+1.36%
Basic ferrous metal manufacturing	3	7.33	−67	+272	+3.72%
Cement, lime, plaster and concrete product manufacturing	4	3.40	−24	−95	−2.80%
Rail freight transport	5	2.87	−29	+16	+0.55%
Fertiliser and pesticide manufacturing	3	2.46	+387	−97	−3.93%
Air and space transport	1	2.10	+209	−96	−4.58%
Petroleum and coal product manufacturing	2	1.77	+6	−68	−3.83%
Other basic chemical product manufacturing	3	1.32	−63	−284	−21.54%
Electricity generation	4	1.19	+117	+26	+2.19%
Gas supply	4	0.81	−23	−54	−6.61%
Basic chemical manufacturing	3	0.67	+54	−34	−5.03%
Pulp, paper and paperboard manufacturing	2	0.41	−29	−29	−7.09%
Grain mill and cereal product manufacturing	1	0.40	+19	−15	−3.85%
Pipeline and other transport	2	0.38	+5	−30	−7.85%
Other non-metallic mineral product manufacturing	1	0.24	+25	−4	−1.52%
Glass and glass product manufacturing	1	0.19	−23	−32	−17.10%
Water passenger transport	1	0.17	−9	−1	−0.90%
Water supply, sewerage and drainage services	1	0.14	+1	+2	+1.37%
Rail passenger transport	1	0.14	+15	−2	−1.75%
Basic ferrous product manufacturing	1	0.11	−2	−4	−3.83%
Road freight transport	1	0.11	+10	−3	−2.66%
Total	163	115.96	−98	+17	+0.01%

Industries in descending order of base emissions; industries with fewer than 3 facilities are still listed, but results for industries represented by only one or two facilities should not be interpreted as industry-wide estimates. A positive relative-intensity effect means worsening relative to the facility's own benchmark. The underlying working paper contains company-specific matching and classification assessments and can be provided confidentially to DCCEE on request.

Table G.2 — 17 Queensland coal mines, covered emissions and measured production, FY2016-17 to FY2024-25

Financial year	Covered, Mt	Gross Raw Output, Mt	Intensity, t CO ₂ -e / t	Relative to FY2016-17
2016-17	6.69	144.4	0.04633	—
2017-18	7.70	152.6	0.05047	+9.0%
2018-19	7.61	142.2	0.05348	+15.4%
2019-20	8.49	143.0	0.05942	+28.3%
2020-21	8.45	136.2	0.06201	+33.9%

Financial year	Covered, Mt	Gross Raw Output, Mt	Intensity, t CO ₂ -e / t	Relative to FY2016-17
2021-22	7.84	134.0	0.05849	+26.3%
2022-23	8.21	138.0	0.05948	+28.4%
2023-24	8.04	131.0	0.06142	+32.6%
2024-25	7.98	133.4	0.05979	+29.1%

The nine-year emissions panel is taken from the Regulator's published tables for each year^[31] and production from the Queensland Department of Resources^[29]. Aggregate emissions intensity increased in six of the eight annual intervals; mine-by-mine annual figures remain noisy, with 136 observations across 17 mines and 8 years, a median annual change of +5.2% and quartiles of –4.0% to +16.4%. The aggregate nine-year trend is clear, but annual results for individual mines are too volatile to support broader extrapolation.

Annex H — References and limitations of the evidence base

Every numbered marker in the text, for example ^[1], points to the corresponding entry below. Quantities marked + are arithmetic on published figures; the main equations are in Annex A.

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Data licences. The Clean Energy Regulator's register, TEBA table, intensity-determination and electricity-sector data, the Office of the Chief Economist forecast tables, the DCCEEW chart data, the Queensland Department of Resources production statistics, the Geoscience Australia mines dataset and the National Pollutant Inventory extract are used and reproduced in the accompanying workbook under the Creative Commons Attribution licences stated on the respective sites. The Global Energy Monitor tracker extracts are used under the Creative Commons Attribution 4.0 International licence and have been filtered and matched as described in Annexes A.5 and A.8. NASA POWER data are in the public domain. International Energy Agency figures are cited, not reproduced.

H.2 Limitations of the evidence base

Jurisdictional coverage. Coal closures were matched from the tracker and checked one by one, relying mainly on the public mine-site data of Queensland and New South Wales. Non-coal facilities were searched through company, administrator and regulatory announcements (September 2026), giving four confirmed closures. The search cannot establish that the list of closures is exhaustive.

Availability of records. The Regulator does not publish stable facility identifiers or facility-level production. Identity matching and production checks therefore rely on National Pollutant Inventory site numbers and state mine-site data, and 61 candidate matches still await manual review.

Inputs not available from public sources. The SMC issuance assumption for below-threshold facilities, the post-2030 TEBA uptake rate, the size of the reserve, and the reserve treatment of the published estimates (other than RepuTex) are not public. No estimates are assigned to them.

Limitations of the modelled series. The full-compliance assumption reduces the modelled base-year result by approximately 1.7 Mt relative to the reported outcome, because the mechanism allows deferral[†]. The gradual entry of the pipeline on the Annex A.5 profile puts the modelled series 1–3 Mt below the Department's projection in FY2026–27 to FY2028–29[†]. The minimum baseline is applied only to facilities with a base-year baseline of at least 50,000 t. These assumptions explain the remaining differences between the modelled series and DCCEEW's published projections. Facility-level workings for each of these items are available to the Department on request, and the accompanying workbook reproduces every table and figure with its formulas so that each difference can be traced to a stated assumption.

Data currency. The register data are the Regulator's FY2024–25 publication; the OCE volumes are the June 2026 forecast tables; the Global Energy Monitor extracts are the May 2026 (coal), September 2025 (LNG) and August 2025 (iron ore) editions; the TEBA table, intensity determinations and electricity-sector data were accessed in August 2026; the NASA POWER series were retrieved on 13 September 2026; and the planning and referral documents cited for the data-centre projects were accessed on 14 September 2026. Later editions of any of these sources may move individual figures.

Corrections. Any material correction identified after submission will be notified to DCCEEW and recorded in a dated correction notice available from Heliovolcan, stating the original figure, the corrected figure, the source relied on and the date of the change.